
NATIONAL ENERGY REGULATOR OF SOUTH AFRICA

In the matter regarding

Concurrence with the ministerial determination on the procurement of 3000 MW from gas technology

By

THE DEPARTMENT OF MINERAL RESOURCES AND ENERGY (DMRE)

DECISION

Based on the available information and the analysis conducted on the draft determination submitted by the Minister of Mineral Resources and Energy in terms of section 34(1) of the Electricity Regulation Act, 2006 (Act No 4 of 2006) ('the Act') and the Electricity Regulations on New Generation Capacity [published as GNR. 399 in Government Gazette No. 34262 dated 4 May 2011, as amended ('the Regulations')], at its meeting held on 14 December 2022, the Energy Regulator decided **to not concur** with the determination that:

1. 3 000 megawatts (MW) of gas power new generation facility at Richards Bay is needed to be procured to contribute towards energy security,
2. the electricity procured shall target connection to the grid as soon as reasonably possible and not beyond the year 2028;
3. there must be prior agreement by NERSA on the cost recovery mechanism.
4. the generation capacity must be procured by Eskom Holdings SOC Ltd through one or more tendering procedures which is fair, equitable, transparent, competitive and cost effective.
5. procurement of primary energy and its associated logistics infrastructure must be done in a manner supportive of the State's participation through other State Owned Companies which have the relevant mandate(s)

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DEFINITIONS

In this Reasons for Decision (RfD) document, any word or expression to which a meaning has been assigned, shall have a meaning so assigned, unless the context indicates otherwise.

Buyer	means, in relation to a new generation capacity project, any organ of state designated by the Minister of Mineral Resources and Energy in terms of section 34(1)(c) and (d) of the Electricity Regulation Act, 2006 (Act No. 4 of 2006). In this regard the buyer is Eskom.
Eskom	means Eskom Holdings SOC Limited, as contemplated in section 3(1) of the Eskom Conversion Act, 2001 (Act No. 13 of 2001).
Government	means the Government of the Republic of South Africa
Independent Power Producer (IPP)	means any person in which the Government or any organ of state does not hold a controlling ownership interest (whether directly or indirectly), which undertakes or intends to undertake the development of new generation pursuant to a determination made by the Minister of Mineral Resources and Energy in terms of section 34(1) of the Electricity Regulation Act, 2006 (Act No. 4 of 2006).
Minister	means the Minister of Mineral Resources and Energy
Organ of state	bears the meaning ascribed to it in section 239 of the Constitution.
Peaking project	means the energy plants or power stations that have very low capacity factors, i.e. generally produce energy for limited periods, specifically during peak-demand periods, with storage that supports energy on demand.
Power purchase agreement	or ' PPA ' means an agreement concluded between a generator and the buyer for the sale and purchase of new electricity generation capacity or electricity derived therefrom, or both.

Procurer	means the person designated by the Minister in terms of section 34 of the Electricity Regulation Act, 2006 (Act No. 4 of 2006) as being responsible for the preparation, management and implementation of the activities related to procurement of new generation capacity under an IPP procurement programme, including the negotiation of the applicable power purchase agreements, which person may or may not be a buyer.
The Act	means the Electricity Regulation Act, 2006 (Act No. 4 of 2006).
New generation capacity	means: (a) the electricity generation capacity other than the capacity of existing generation facilities; (b) the electricity derived from the capacity referred to in (a); and (c) ancillary services related thereto, individually or in any combination thereof, and including an increase in the electricity generation capacity of existing generation facilities.

ABBREVIATIONS AND ACRONYMS

CO	Commercial Operation
CCGT	Combined Cycle Gas Turbine
DMRE	Department of Mineral Resources and Energy
DPE	Department of Public Enterprises
EIUG	Energy Intensive User Group
ESI	Electricity Supply Industry
FDI	Foreign Direct Investment
FY	Financial Year
GDP	Gross Domestic Product
ICE	Internal Combustion Engine
IRP	Integrated Resource Plan
IPP	Independent Power Producer
ISMO	Independent System and Market Operator
LNG	Liquid Natural Gas
LPU	Large Power Users
MES	Minimum Emission Standard
kW	Kilowatt
MW	Megawatt
NERSA	National Energy Regulator of South Africa
OCGT	Open Cycle Gas Turbine
PAJA	Promotion of Administrative Justice Act
PPA	Power Purchase Agreement
RCA	Regulatory Clearing Account
RE	Renewable Energy
REIPPPP	Renewable Energy Independent Power Producer Procurement Programme
RfD	Reasons for Decision
RFI	Request for Information
RFP	Request for Proposal
RMIPPPP	Risk Mitigation Independent Power Producer Procurement Programme

TS	Transmission System
TNSP	Transmission Network Service Provider
UCLF	Unplanned Capability Loss Factor

1. INTRODUCTION

- 1.1 The National Energy Regulator of South Africa (NERSA) is a regulatory authority established as a juristic person in terms of section 3 of the National Energy Regulator Act, 2004 (Act No. 40 of 2004). NERSA's mandate includes regulation of the Electricity Supply Industry.
- 1.2 In accordance with section 34 of the Electricity Regulation Act, 2006 (Act No. 4 of 2006) ('the Act'), the Minister of Mineral Resources and Energy ('the Minister') may, in consultation with the Energy Regulator:
- a) determine that new generation capacity is needed to ensure the continued uninterrupted supply of electricity;
 - b) determine the types of energy sources from which electricity must be generated, and the percentages of electricity that must be generated from such sources;
 - c) determine that the electricity thus produced may only be sold to the persons or in a manner set out in such notice;
 - d) determine that electricity thus produced must be purchased by the persons set out in such notice; and
 - e) require that new generation capacity must:
 - i. be established through a tendering procedure that is fair, equitable, transparent, competitive and cost-effective, and
 - ii. provide for private sector participation.
- 1.3 The proposed section 34 determination is based on an application made by Eskom on 13 January 2022 to the Minister for the development of a 3 000 MW Combined Cycle Gas Power Plant at Richards Bay.

2. LEGAL MANDATE

2.1 NERSA legal mandate

- 2.3.1 Section 4(c) of the National Energy Regulator Act, 2004 empowers NERSA with the responsibility to undertake the functions detailed in section 3 and section 4 of the Electricity Regulation Act, 2006.
- 2.3.1 The Act sets out the powers and functions of NERSA. Section 34 of the Act allocates the powers to initiate new generation capacity. NERSA does not have

the power to initiate the need for new generation capacity; however, the process to initiate new generation capacity cannot move forward without approval by NERSA. The section re-emphasises the powers of NERSA as the custodian of the regulatory framework. It must also be clear from the onset that section 34 does not delegate the powers to NERSA, similar to the section 4 mandate.

- 2.3.1 Section 34 designates the Minister as the authority who must generate the determination. The powers of the Minister are not without a proviso. The section goes further to demand that, in order to implement the determination decision, such exercise must be in consultation with NERSA. The 'in consultation' principle carries the power to the finalisation of the draft determination.
- 2.3.1 The Act does not expand on how NERSA should go about meeting the dictates of section 34. The silence of the section on how NERSA should address the dictates of section 34 is pivoted on the existing application of the 'in consultation' principle. In consultation demands that the concurrence of NERSA must be sought by the Minister before exercising a delegated power. This known legal principle has generated a mandate to NERSA to process the determination.
- 2.3.1 The absence of concurrence by NERSA with regard to the draft determination renders the proposition by the Minister unachievable. It is also clear that when NERSA concurs, consideration is given with regard to what has been provided in the draft determination and within the provisions of section 34.
- 2.3.1 In performing its mandated functions, NERSA is required to ensure that the following objects set out in section 2 of the Act are achieved:
- a) The efficient, effective, sustainable and orderly development and operation of electricity supply infrastructure in South Africa.
 - b) That the interests and needs of present and future electricity customers and end-users are safeguarded and met, having regard to the governance, efficiency, effectiveness and long-term sustainability of the electricity supply industry within the broader context of economic energy regulation in the Republic.
 - c) That investment in the electricity supply industry is facilitated.
 - d) That universal access to electricity is facilitated.

- e) That the use of diverse energy sources and energy efficiency is promoted.
- f) That competitiveness and customer and end-user choice are promoted.

2.3.1 The objects of the Act define the outcomes that all actions and decisions undertaken by NERSA in respect of its functions seek to achieve. In concurring with the Minister's section 34 determination, NERSA gauges how closely the determination aligns with the objects of the act.

2.3.1 The process that NERSA has to undertake in concurring with the Minister's determination must satisfy the dictates of section 10(g) of the National Energy Regulator Act and the provisions of the Promotion of Administrative Justice Act, 2000 (Act No.3 of 2000) ('PAJA'), to render the decision-making process lawful.

3. BACKGROUND

3.1 The Applicant

3.1.1 The proposed section 34 determination is based on an application made by Eskom on 13 January 2022 to the Minister of Mineral Resources and Energy ('the Minister') for the development of a 3 000 MW Combined Cycle Gas Power Plant in Richards Bay.

3.1.2 During May 2022, the Department of Mineral Resources and Energy (DMRE) requested that Eskom submit additional information on the technical and commercial aspects of the project. Eskom then proceeded to supplement its application with all the information requested by the DMRE.

3.1.3 The project has progressed; Eskom is currently in the project definition phase while environmental approvals for both the power plant and the Transmission Grid connection have been obtained.

Need for flexible generation capacity to support RE

- 3.1.4 On 25 July 2022, His Excellency President Cyril Ramaphosa announced measures to be taken to address the current energy crisis, which included increasing capacity to be procured under the Renewable Energy Independent Power Producer Procurement Programme (REIPPPP). The REIPPPP was aimed at bringing additional megawatts onto the country's electricity grid through private sector investment in wind, solar photovoltaic (PV), storage and other technologies.
- 3.1.5 Gas-to-power technologies in the form of Combined Cycle Gas Turbine (CCGT), Closed Cycle Gas Engine (CCGE) or Internal Combustion Engine (ICE) provide the flexibility required to complement renewable energy. In the short term, the goal is to pursue gas import options, as local and regional gas resources will allow for scaling up within manageable risk levels. Exploration to assess the magnitude of local recoverable shale and coastal gas are being pursued and may be accelerated.
- 3.1.6 There is enormous potential and opportunity to develop a Gas Market in South Africa and the Brulpadda gas resource discovery in the Outeniqua Basin of South Africa, piped natural gas from Mozambique (Rovuma Basin), indigenous gas like coal-bed methane and ultimately shale gas, could form a central part of South Africa's strategy for regional economic integration within the Southern African Development Community (SADC).
- 3.1.7 Cooperation with neighbouring countries is being pursued and partnerships are being developed for joint exploitation and beneficiation of natural gas within the SADC region. The SADC is developing a Gas Master Plan to identify the short- and long-term infrastructure requirements to enable the uptake of a natural gas market. Availability of gas would also provide an opportunity to convert to CCGT and run open-cycle gas turbine plants at Ankerlig (Saldanha Bay), Gourikwa (Mossel Bay), Avon (outside Durban) and Dedisa (Coega IDZ) on gas.
- 3.1.8 On 28 July 2022, NERSA received a letter with a section 34 determination from the DMRE, requesting NERSA to confirm concurrence with the section 34 determination for the procurement of 3000 MW from Gas.

3.1.9 According to the determination, the procurement programme shall target connection to the Grid for the new generation capacity as soon as reasonably possible. The determination is detailed below and attached hereto as **ANNEXURE A**.

DETERMINATION UNDER SECTION 34(1) OF THE ELECTRICITY REGULATION ACT, 2006 (ACT NO. 4 OF 2006)

3.1.10 The Minister, in consultation with NERSA, acting under section 34(1) of the Act, has determined that 3000 MW should be procured, to be generated from Gas as described below:

- a) the electricity procured shall target connection to the grid as soon as reasonably possible and not beyond the year 2028;
- b) there must be prior agreement by NERSA on the cost recovery mechanism;
- c) the generation capacity must be procured by Eskom Holdings SOC Ltd through one or more tendering procedures that are fair, equitable, transparent, competitive and cost effective; and
- d) the procurement of primary energy and its associated logistics infrastructure must be done in a manner supportive of the State's participation through other State Owned Companies that have the relevant mandate(s).

4. THE NERSA DECISION-MAKING PROCESS

4.1 Process followed by NERSA to reach the decision

4.1.1 NERSA received the request to concur with the section 34 determination on 28 July 2022 for the procurement of 3000 MW to be generated from gas.

4.1.2 On 24 August 2022, NERSA's Electricity Subcommittee (ELS) approved the NERSA Consultation Paper on the proposed procurement of the new generation capacity.

4.1.3 On 25 August 2022, NERSA published the consultation paper, requesting stakeholders to submit written comments to enable NERSA to appropriately apply its regulatory reviews and decision making prior to concurrence with the Minister.

4.1.4 The closing date for the submission of comments was 16 September 2022.

4.1.5 NERSA received substantive comments from stakeholders, which were largely supportive of the determination received from the Minister.

5. STAKEHOLDER COMMENTS

5.1 Comments received from stakeholders

5.1.1 NERSA received substantive comments from stakeholders, which were supportive of the determination received from the Minister.

5.1.2 The comments were taken into consideration during the consideration of the determination and are also recognised in this Reasons for Decision (RfD) document. The written comments, as well as NERSA's analysis thereof, is attached as Annexure B.

5.1.3 Below is a list of all stakeholders who submitted written comments:

- a) Business Unity South Africa (BUSA)
- b) Centre for Environmental Rights (CER), Natural Justice (NJ), Groundwork and South Durban Community Environmental Alliance (SDCEA)
- c) Eskom
- d) Energy Intensive Users Group (EIUG)
- e) Gigajoule International
- f) Meridian Economics
- g) Minerals Council South Africa
- h) Moondream
- i) Phakwe Group
- j) Richards Bay Industrial Development Zone (RBIDZ)
- k) SASOL
- l) Umhlathuze City Development.

Stakeholder comments

- 5.1.4 In general, stakeholders supported the ministerial determination for the procurement of 3 000MW of new generation capacity from gas with the exception of a few stakeholders who rejected the proposal based on environmental concerns.
- 5.1.5 The Centre for Environmental Rights (CER), Natural Justice (NJ), Groundwork and South Durban Community Environmental Alliance (SDCEA) raised their objections to the ministerial determination, stating that the draft gas determination lacks key information and is not (to their knowledge) informed by any electricity needs and least-cost planning or crucial impact assessments, and is thus irrational and unreasonable.
- 5.1.6 The CER further highlighted that the consultation process had been inadequate, and that the proposed determination was in conflict with the law and constitution in that developing new gas will pose unjustifiable harms to the people of South Africa. The determination has also been tailored and retrofitted to an existing, authorised Eskom gas-to-power project.
- 5.1.7 As a proposer of the new generation capacity, Eskom supports the determination for the development of the new 3 000MW CCGT gas project, but raised a concern that the Draft Determination from the Minister does not accurately reflect Eskom's intentions for the project, as stated in its request for a section 34 determination submitted to the Minister.
- 5.1.8 The EUIG supports the determination, stating that gas is a transitional power source towards net-zero goal. For EUIG members who are base-load electricity consumers, gas is a reliable source due to its dispatchability, especially where intermittent renewable energy is not available. The 3 000MW of capacity will assist in ensuring security of supply, however it is not considered sufficient to ensure security of supply on its own.

- 5.1.9 Gigajoule supports the determination in principle, but stated that in light of South Africa's current energy outlook, it would be good practice to take a portfolio approach to the gas-to-power make-up of the country. There are options that will provide the same or better results with an improved risk profile and positive downstream value addition, compared to 'placing all the eggs in one Richards Bay basket'.
- 5.1.10 Meridian Economics objected to the ministerial determination, citing concerns that based on their findings, it is not economically rational to commit to large-scale gas use power in this decade and following such a strategy would deliver assets that are stranded before their first kWh of power has been generated. Their analysis found that the economic role of gas in the South African power sector is small and its viability depends on 'anchor demand' from other sectors.
- 5.1.11 The Minerals Council of South Africa is mainly concerned with affordability since the gas must be procured in US dollars. They state that with the Rand being volatile, price certainty will be difficult. The price of liquid fuels must be considered to understand the extent of the volatility that the electricity market will be subjected to because of having to import the gas.
- 5.1.12 Moondream supports the ministerial determination, citing the fact that Natural Gas-fired power plants enjoy a competitive position in terms of quick start-up and switching-off, making it the best available technology to aid the integration and use of renewable energy to the grid.
- 5.1.13 Phakwe Group supports the determination and said that Gas will allow for the acceleration of decommissioning of coal plants, while also allowing for the increased use of renewables. The group further stated that Gas will cover the current energy gap, thus potentially reducing the possibility of load-shedding.
- 5.1.14 The RBIDZ fully supports the proposal to issue a section 34 determination to Eskom Holdings SOC Ltd to allow it to build a 3 000 MW gas power plant in Richards Bay. The Board of the RBIDZ has already approved the allocation of a 68 ha site in Phase 1D of the RBIDZ to Eskom.

5.1.15 Sasol supports the broader intent and principles contained in the determination and NERSA's consultation paper stating that Gas, together with renewable energy (RE), are both cleaner technologies than coal in the transition to a lower carbon economy and South Africa's commitment towards net zero.

5.1.16 The City of Umhlathuze supported the determination in principle, however emphasised that the concerns raised by interested and affected parties should be considered carefully and NERSA should ensure that all legislative requirements are met while concluding on this section 34 determination.

5.1.17 Overall, stakeholders agreed that the determined capacity is required, but also emphasised the need to perform a proper techno-economic power system analysis, a 'value for money' analysis (as required by the New Generation Regulations), and a 'cost-effectiveness' test as required by section 217(1) of the Constitution. These are to be conducted by Eskom, the Minister of Mineral Resources and Energy or NERSA.

6. DEMAND AND SUPPLY SIDE ANALYSIS

6.1 Eskom's demand and supply side motivation for the development of 3 000MW from gas power technology in the form of CCGT

6.1.1 The electricity system has been constrained for most of the past 14 years due to the aging and unreliable coal-fired plant fleet. As a result, the assumed Energy Availability Factor (EAF) of 71% in 2020 that was used in the IRP2019 assumptions, decreasing to 72.5% in 2021, has not been achieved.

6.1.2 In fact, by mid-November 2021, the EAF dropped significantly and by the end of the financial year (i.e. March 2022), the EAF was 63.7%. The estimations used by Eskom based on the EAF of 63.7% predicted a further 1% reduction in the EAF, which resulted in a 500MW negative system impact. This means a negative impact of approximately 4 400MW in relation to the assumptions that were used during the development of the IRP2019.

6.1.3 Eskom is therefore applying to develop 3 000MW from gas to power technology in the form of CCGT in order to alleviate the current supply–demand imbalance in the country. This is also in light of the outdated EAF assumptions that are listed in the promulgated IRP2019.

Supply Side Analysis

- 6.1.4 South Africa has been suffering from capacity shortages since 2008 when the country experienced its first bouts of load-shedding. In the course of time, the Eskom coal plants, which meet the majority of South Africa's total demand (95%), began to buckle under the pressure of extensive usage and reduction in maintenance due to the pressure imposed on Eskom to keep the lights on at all cost.
- 6.1.5 This led to the EAF of the coal fleet decreasing steadily from around 80% in 2010 until it has reached approximately 55% in FY 2022/23, which is a decrease of nearly 30% in the last two decades. It was assumed in the IRP2019 that the EAF would rise to 73.5% in FY 2022/23, however this has not materialised due to various reasons, including over-usage of the coal units, lack of proper reliability maintenance, and a lack of adequate finances to conduct maintenance, among other issues. The power shortages that have led to load-shedding were also exacerbated partly by the design problems and delays to the two new coal-fired megaprojects, i.e. Medupi and Kusile.
- 6.1.6 In March 2021, Kusile Power Station's third unit attained commercial operation (CO), which brought an additional 720 MW to the system. At Medupi Power Station, all six of the 720 MW units are already online, except for unit 4 whose Generator suffered a major failure, removing 720 MW of baseload capacity from the grid.
- 6.1.7 South Africa is planning to close its capacity shortage gap by procuring approximately 15 GW of new generation capacity by 2030 as per the Integrated Resource Plan (IRP 2019). The first part of that capacity is being procured under the Risk Mitigation Independent Power Producer Procurement Programme (RMIPPPP), which seeks to obtain 2 GW of firm and dispatchable capacity based on a competitive bidding process, with the aim of going online by the end of 2022.

- 6.1.8 The preferred bidders for the RMIPPP program were announced in March 2021. More than half of the allocated capacity (1.15 GW) of the program will come from three Liquid Natural Gas (LNG) power ships at different locations on the coastline. However, this increase in gas generation in the supply mix, as per the current section 34 determination, will depend on the successful rollout of LNG infrastructure.
- 6.1.9 In addition, several renewable plants from Bid Window 5 of the Renewable Energy Independent Power Producer Procurement Programme (REIPPPP) came online in late 2020 and early 2021, while further projects from this bid window which were expected to come online in 2022 have been delayed due to Covid-19 resulting in a number of them failing to reach financial close.
- 6.1.10 Despite the new capacity, the fortunes of the South African power system have not improved, with significant load-shedding which occurred in July 2022 reaching an unprecedented level 6, caused by a deficit of 6 000MW. The power deficit of 4 000MW to 6000 MW also results in the increased use of Open Cycle Gas Turbines (OCGT) which burn expensive diesel.
- 6.1.11 To close the energy gap, Eskom has several options available to it. The first option is to increase the use of Open Cycle Gas Turbines (OCGTs), which have proved to be capable but expensive. Figure 1 below shows the increased usage of OCGTs particularly during the winter months. The second option is to embark on an intensive maintenance program to improve the performance of the existing Eskom coal fleet, thus improving the Energy Availability Factor (EAF). The third option that is currently being pursued is to build new generation capacity by adding new MWs to the grid.

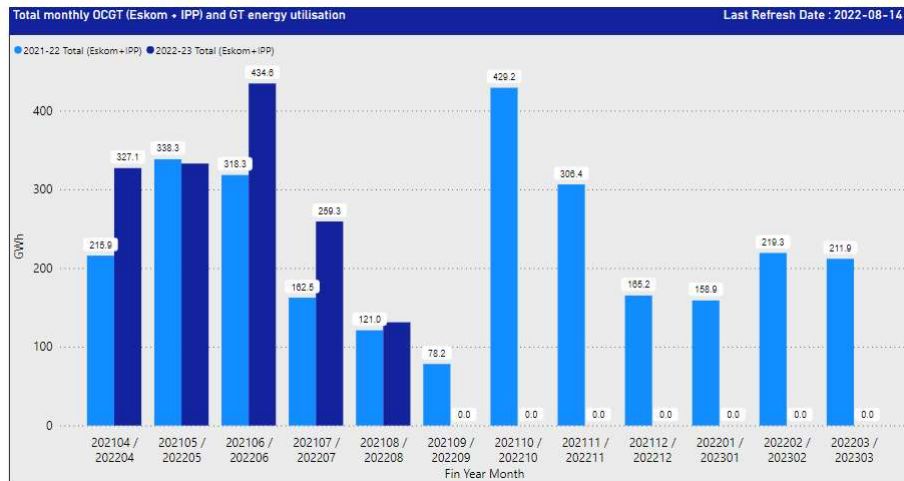


Figure 1: Monthly usage of OCGTs in South Africa
Source: Eskom (2022)

6.1.12 Among the options available to Eskom to balance supply and demand, load-shedding has the worst impact on the economy because it results in customers deflecting by adopting alternative energy sources such as roof-top solar. An unreliable energy supply also results in the reduction of sales volumes by Eskom exacerbating the possibility of Eskom entering the utility death spiral.

6.1.13 Load-shedding also leads to loss of production resulting in job losses. Furthermore, according to Alexforbes (2022) load-shedding costs the economy approximately R4 Billion a day due to unserved energy. Figure 2 below depicts load-shedding in South Africa since 2015.

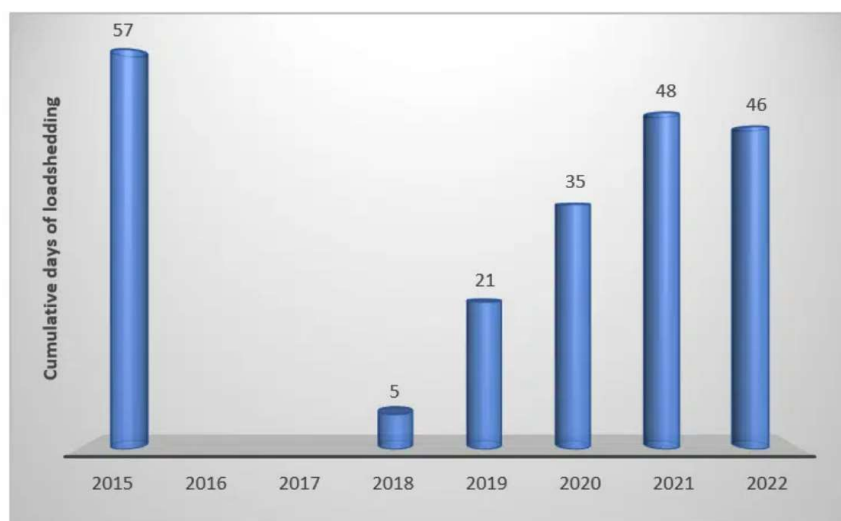


Figure 2: Load-shedding in South Africa since 2015
Source: My broadband (2022)

- 6.1.14 Figure 2 above indicates that 2016 and 2017 were good years for Eskom where there was no load-shedding for the country and the EAF improved by about 6%. However, from 2018, the performance of the coal fleet entered a new phase of continued deterioration in performance, which could have been as a result of accelerated aging and poor quality maintenance, among other issues that are well documented in literature.
- 6.1.15 According to the Medium-term system adequacy outlook (MTSAO) released by the System Operator (SO) in October 2021, demand currently exceeds supply by at least 4000MW to 6000MW. The proposed capacity of 3 000 MW from Gas will therefore contribute towards reducing the risk of load-shedding and reduce diesel generator usage. Furthermore, the capacity of 3 000 MW from Gas will also be a useful lever to restore system adequacy and improve security of supply while effort is being made to improve Eskom's coal plant EAF.
- 6.1.16 Finally, decision 7 of the IRP2019 states that, to support the development of gas infrastructure and in addition to the new gas to power capacity in Table 5, convert existing diesel-fired power plants (Peakers) to gas. This determination could present an opportunity to execute on decision 7 of the IRP. However, proper feasibility studies will reveal the most efficient course of action to realise this much needed new generation capacity.

Demand Analysis

- 6.1.17 Electricity demand is forecasted to remain below 2019 levels on account of stifled demand as certain sectors struggle to operate in the current climate of electricity shortages and due to the impact of Covid-19 lockdowns. This lower demand is a window of opportunity to close the 6000 MW capacity gap. However, the power shortages are expected to continue until new generation comes online by the end of 2022 at the earliest and more likely in 2023, under the RMIPPPP program.

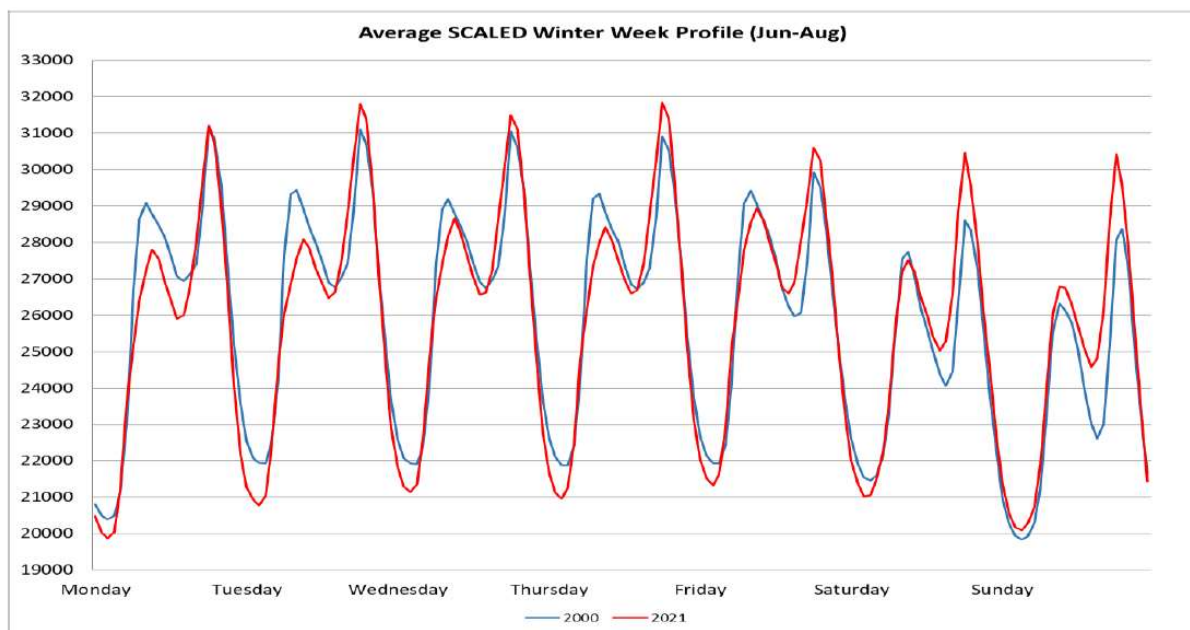


Figure 3: Demand profile in South Africa 2021 compared to year 2000
Source: Eskom

6.1.18 Figure 3 above is a snapshot of the demand profile taken over an arbitrary winter week where demand is the highest in the year 2000 compared to year 2021. The profiles show that the demand for electricity in South Africa have remained stifled over the last two decades, even declining slightly.

6.1.19 This is due to a number of economic and socio-economic factors including declining economic growth, lack of generation capacity and higher prices of electricity which have increased by more than 460% between 2007 and 2020 not accounting for inflation.

6.1.20 Figure 4 below depicts the South African Grid Residual Demand for 2015 when the RE penetration was at 2 021MW and the demand shape of 2019 when the RE penetration was at 3 981MW. The required ramp rate in 2015 was approximately 0.5MW/pm, the required ramp rate in 2019 was approximately 0.8MW/pm, resulting in a 60% increase in ramp rate requirement. This impact will worsen as more RE is introduced into the South African energy mix. Ramping coal plants up and down in this manner further accelerates their aging.

6.1.21 Since the Registration limit has been increased from 1MW to 100 MW, and it is envisaged that most of the new embedded generation capacity will be PV plants that all fall away at evening peak, this will result in an even steeper peak at national grid level. To counter this effect, a significant amount of fast peaking generation i.e., Gas, needs to be proportionally incremented to prevent the duck curve phenomenon shown in Figure 4 below.

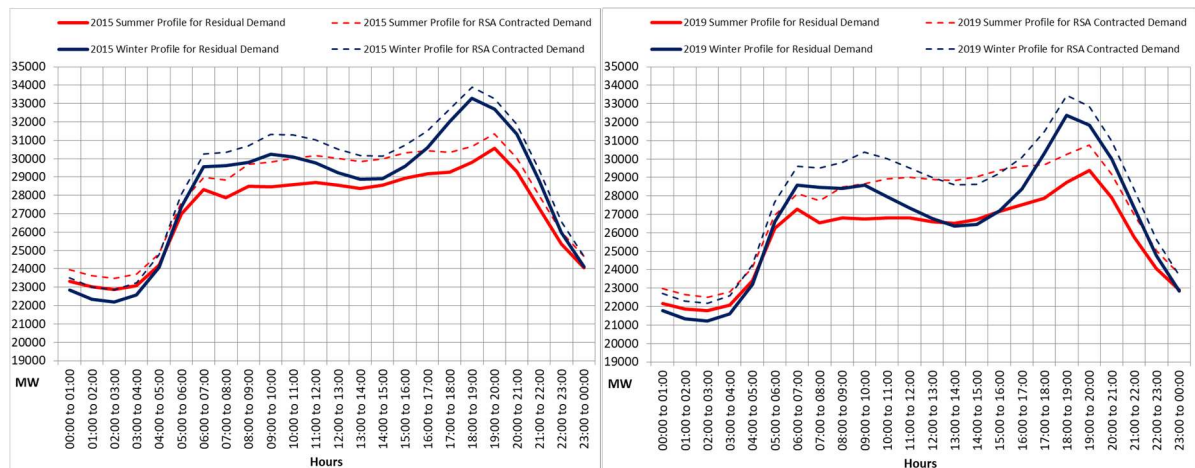


Figure 4: Typical Day Residual Demand Profile in RSA Grid comparing 2015 and 2019 residual demand shape

6.1.22 The duck curve presents significant challenges for conventional power generation technologies that have to balance out the system, since many of them are not able to ramp up quickly enough. The term “duck curve” is used as the load profile resembles the silhouette of a duck. This phenomenon poses the following challenges;

- PV does not reduce the peak demand because peak demand is predominantly before sunrise and after sunset. This poses challenges for the SO and grid operators as it is difficult to control the grid, with these sharp increases and reductions in load especially the sharp increase at evening peak time.
- This results in the peak being much steeper in the evening and requiring much faster ramping than the conventional generation plants can cope with.
- Given that South Africa has no interconnections with other strong grids as stated above, the only option to deal with the duck curve is an increase in the flexible generation such as natural gas and storage solutions e.g. batteries which are quick to dispatch and can ramp up at the rate that the system would require.
- The steeper the evening peak becomes, the more the system will require standby and reserve capacity to support the evening peak.

- For high levels of PV in the network, the residual demand can run into negative values in the middle of the day. This negative load can cause voltage and stability problems, see Figure 5 below. To counter this phenomenon, battery storage is required to absorb additional energy generated from solar to be discharged during the evening peak. This would help to improve the residual load profile below.

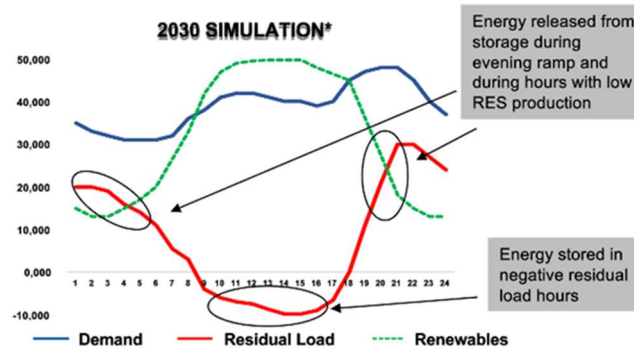


Figure 5 Net load curve with increased VRE penetration

Source: Terna: Italian Transmission System operator

6.1.23 The planning and operation of the transmission and distribution network will be more complex, the purpose of these networks will not only be to feed power to load centres but to control power quality as well. Parameters affected by the introduction of Embedded Generators into a network include:

- Fluctuating voltage levels,
- Higher fault levels,
- Thermal loading,
- Technical losses,
- Stability.

6.1.24 NERSA therefore agrees with stakeholders indicating that Gas is reliable and efficient technology and will contribute to base load and stability of the grid, as well as the fact that security of supply would give rise to security of demand in the medium to long term as industries can access affordable, stable and reliable electricity. To stimulate industrialisation, it is critical to maintain a constant, affordable and reliable source of electricity. This would then attract much needed foreign direct investment (FDI) into the country.

6.1.25 Industrial stakeholders/customers, particularly metallurgical sectors have indicated that the combination of unreliable supply of electricity and the large increased in electricity prices, prohibit the optimisation of their production and impede their economic efficiency, which has an impact on their financial sustainability, their competitiveness in the global economy, and their ability to grow. Given the high levels of unemployment in South Africa, the situation of unreliable supply of electricity should therefore not be allowed to continue.

System needs analysis

6.1.26 The System Operator operates the grid and ensures that at system level there is a balance between supply and demand. South Africa currently has baseload and peaking capacity but lacks mid-merit capacity in the form of gas-to-power. Mid-merit capacity is essential to ensure that the system is able to load-follow i.e., the slight up and down load profile fluctuations throughout the day are currently met with baseload power plants which affects their optimal performance. Load following is best suited for flexible capacity such as gas.

Eskom's motivation for system needs analysis

6.1.27 The Generation Connection Capacity Assessment (GCCA) published by Eskom at the end of October 2021 indicated that there will only be 1 236MW of transmission line capacity available in the Eastern Cape after Bid Window 5 projects from Renewable Energy Independent Power Producer Programme (RPPPP) are connected. Therefore, it would be reasonable to leave the remaining transmission capacity for renewables as they highly depend on favourable natural occurring resources such as wind speed and sun radiance.

6.1.28 Appropriate studies were then conducted by Eskom to make sure that the 3 000MW to be generated from gas power technology in a form of CCGT should be accessible at an appropriate geographical location to prevent it from being islanded due to transmission grid constraints. Therefore, the studies indicated that this capacity will be correctly positioned in the Richards Bay area compared to Coega area in the Eastern Cape.

6.1.29 The benefits of 3 000MW from gas power technology in a form of CCGT especially in Richards Bay Transmission corridor/area are that it can be ramped fast enough to meet a drop in demand when the sun is not shining and the wind is not blowing on the other provinces where renewables are located and where the grid is constrained. Hence, provide benefits of flexibility and distributed generation across the key economic sectors of the country.

6.1.30 Transmission corridor in Richards Bay also provides many benefits because the gas can either be imported via ships or be transported via pipelines because it is within the boundaries of Mozambique gas pipeline and that of Transnet Lilly gas pipeline.

6.1.31 Furthermore, the capacity of 3000MW from Gas will assist the system with flexible capacity by filling in the supply gaps as shown in Figure 6 below, which RE cannot fill due to its intermittency. This will help with the balancing of supply and demand and reduce Eskom's operating costs. The effect will be improved grid stability and improved efficiency.

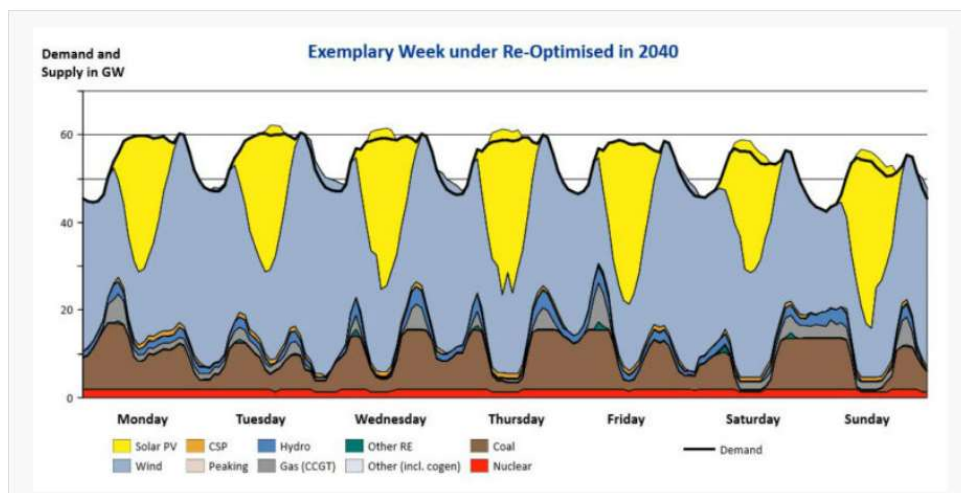


Figure 6: Role of gas to meet total demand
Source: CSIR

Stakeholder comments on the sufficiency of the capacity to be procured under the 3000MW Gas technology Procurement program

6.1.32 Stakeholders stated that Eskom indicated a need for 4000 MW to 6000 MW additional baseload generation capacity and this has not been addressed, thus the supply will not be enough for the demand. They further stated that the capacity of 3000MW CCGT is insufficient in the context of Eskom's deteriorating coal fleet efficiencies and this needs to be addressed by converting some of the older coal generation plants to gas. Other stakeholders stated that the capacity of 3000MW from Gas is not needed at all to ensure uninterrupted supply of electricity. Instead, they stated that renewables, with other flexible generation, can meet the demand currently being proposed for gas.

NERSA analysis of the Stakeholder comments¹

6.1.33 Climate change is one of the greatest challenges of our time but of equal importance, however, is the need to ensure access to energy for quality of life and for economic development. It is therefore critically important to address climate change as part of the sustainable development agenda and there is ongoing progress in the development of new technologies that have brought confidence and hope that these objectives will be met in the energy system in the future.

6.1.34 The so called just transition must be ensured as we transition to clean economies, meaning that the use of diverse energy sources ensures that security of supply is ensured by dealing with risks associated with intermittency of RE capacity as well as high cost of storage, while ensuring that strides are taken towards green economies. It has to be at a pace that takes South African set of circumstances into account.

6.1.35 Dramatic price reductions and technological advancement of wind generators and solar photovoltaics (PV) have shown that these renewable energy resources can be important players in global electricity systems, and that the long-anticipated breakthrough in cost-effective storage technology would shift energy mixes substantially. These developments have invariably led to an assumption that the era of fossil fuels across the energy system is effectively over and that there is no need for further development of new resources and that we have to stop using them as soon as possible.

¹ Based on a write-up by Scott Foster and David Elzinga available at: <https://www.un.org/en/chronicle/article/role-fossil-fuels-sustainable-energy-system>, accessed on 21 September 2022

6.1.36 This assumption has also led to a perception of “good” renewables-based technologies on the one hand, and “bad” fossil fuels-based technologies on the other hand. The reality however is that this debate is much more nuanced and requires more thorough investigation. Further development of technologies that will allow fossil fuels to become "part of the solution", rather than remain "part of the problem" is needed because as seen recently in countries such as Germany², all technologies have a role to play in an energy system that is guided by rational economics. Most stakeholders therefore echo this sentiment in recognition of the realities above.

7. COST RECOVERY MECHANISM

7.1 Cost Analysis

7.1.1 The purpose of the cost analysis is to ensure that the addition of the proposed capacity will ensure that only prudently incurred costs are recovered to guarantee the principle of cost reflectivity and a reasonable return. This will ensure minimum impact on overall electricity tariffs and transparency resulting in least impact on end users.

Stakeholder comments on the cost and financial impact of the new generation capacity

7.1.2 Stakeholders raised the concern of affordability since the gas must be procured in US dollars and that the volatility of the Rand could affect price certainty. This was raised in recognition of the fact that South would largely have to depend imported gas.

7.1.3 Stakeholders further stated that surmounting this problem would be a complex exercise as it involves hedging on two fronts: hedging against the depreciation of the rand against the US dollar; and hedging against the international gas price volatility. The view of stakeholders therefore is that without significant domestic production of gas, price volatility of gas generated electricity may be inevitable.

² Germany reactivates coal-fired power plant to save gas, available at: <https://www.dw.com/en/germany-reactivates-coal-fired-power-plant-to-save-gas>, accessed on 21 September 2022

NERSA Cost Analysis

7.1.4 In its feasibility study, Eskom provided input assumptions for its costs. Eskom's assumptions and results of its nominal Capex were based on Electrical Power Research Institute (EPRI) and Lazard benchmark as indicated in Table 1 below.

Table 1: CCGT Costs based on Lazard and EPRI

R/MWh	LAZARD	EPRI
Component	Base R/MWh	Base R/MWh
Fixed Operating Cost	56	10
Variable O&M Cost	58	49
Fuel Cost	1 236	1 236
Capex	394	234
Total	1744	1 529
Nominal Capex (excluding IDC)	R59.9 billion	R35.6 billion

7.1.5 To obtain the costs associated with the construction, operation and decommissioning of the new generation capacity from Gas, the levelised cost of electricity (LCOE) which measures lifetime costs divided by energy production and calculates present value of the total cost of building and operating a power plant over an assumed lifetime is used. LCOE also allows for the comparison of different technologies (e.g., Wind, Solar, and Natural Gas) of unequal spans, project size, different capital cost, risk, return and capacities. The following breakdown is normally used:

- (a) Overnight costs
- (b) Fuel Costs
- (c) Long-term Fuel Waste Management Costs
- (d) Operation and Maintenance Costs
- (e) Decommissioning and Dismantling Costs

7.1.6 The indicative price in (R/kWh) indicated that consumers are expected to pay above R1, 05/kWh (Current year). Some also indicated that the costs of gas-to-power would be above the blended tariff, but less than the high RE average prices of about R2.20.

NERSA Analysis of Cost of Gas-to-power

7.1.7 NERSA notes the concerns raised regarding the high capital costs of gas-to-power technology as well as the tendency for projects built by state-owned-enterprises (SOEs) to be prone to delays as well as costs over-runs. This could however be mitigated by extensive upfront planning and sufficient time being allocated to design and following an appropriate procurement process to mitigate delays and cost overruns.

7.1.8 The cost of gas is determined by the following elements:

- (a) capital costs of construction (including finance costs),
- (b) fuel costs,
- (c) operation and maintenance (O&M) costs, and
- (d) decommissioning costs.

Recent International Published Cost Data

7.1.9 Analysis of the international publicly available data is based on *Projected Costs of Generating Electricity – 2020 Edition (9th Edition)*, which discusses the levelised costs of generating electricity (LCOE). The basis for using this report is that it is produced every five years by the International Energy Association (IEA) and the OECD Nuclear Energy Agency (NEA) under the oversight of the Expert Group on Electricity Generating Costs (EGC Expert Group). It publishes plant-level costs of generating electricity from various regions and countries for both baseload electricity generated from fossil fuels and nuclear power stations, and a range of renewable energy generation.

7.1.10 Table 2 below shows the published capital costs of CCGT in 2020. From the table it can be seen that the overnight costs or investment cost of CCGT technologies varies between US\$560 (in China) and US\$1 109 (in Japan). This reflects a wide range of the overnight costs.

Table 2: Overnight and Investment costs of CCGT

Table 3.2a: Combined-cycle gas turbine (CCGT) generating technology							
Country	Technology	Net capacity (MWe)	Electrical conversion efficiency (%)	Overnight costs (USD/kWe)	Investment costs (USD/kWe)		
					3%	7%	10%
Australia	CCGT	506	48%	955	998	1 058	1 105
	CCGT (CCUS)	437	41%	2 826	2 955	3 133	3 271
Belgium	CCGT	500	60%	767	802	850	888
	CCGT	500	64%	1 009	1 055	1 119	1 168
	CCGT	500	58%	974	1 018	1 079	1 127
Canada	CCGT	471	57%	1 058	1 106	1 173	1 224
Italy	CCGT	790	60%	590	617	654	683
Japan	CCGT	1 372	56%	1 109	1 160	1 229	1 283
Korea	CCGT	481	56%	1 107	1 158	1 228	1 281
	CCGT	982	59%	838	877	929	970
Mexico	CCGT	503	51%	668	700	742	775
	CCGT	785	60%	667	698	740	772
	CCGT	835	58%	466	488	517	540
Romania	CCGT	750	58%	254	265	281	294
United States	CCGT	727	59%	852	895	1 055	1 102
	CCGT (CCUS)	646	48%	2 412	2 522	2 674	2 791
Non-OECD countries							
Brazil	CCGT	980	58%	958	1 002	1 082	1 108
China	CCGT	475	58%	560	586	621	648

Source: EIA 2020 report for different countries and technologies

7.1.11 Figure 7 shows the plots indicating the range of investment costs and fixed O&M costs of various technologies, including OCGT and CCGT (see red arrows), as published in the IEA 2020 report. Also shown are the averages and medians of the investment costs and fixed O&M costs.

7.1.12 The mean values for investment and fixed O&M costs for gas technologies are about \$1050/kW for CCGT and about \$70/kW for OCGT respectively. This is much cheaper than nuclear and coal whose investment costs are higher but have lower operating costs than gas due to their cheaper fuel costs.

7.1.13 Figure 8 below shows the box plot of LCOE value from the IEA 2020 report, calculated at 7% discount rate for various power generation technologies. As can be seen in the plot, the median of gas (CCGT) lies within the distribution of nuclear technology, but lower than the distribution of Coal.

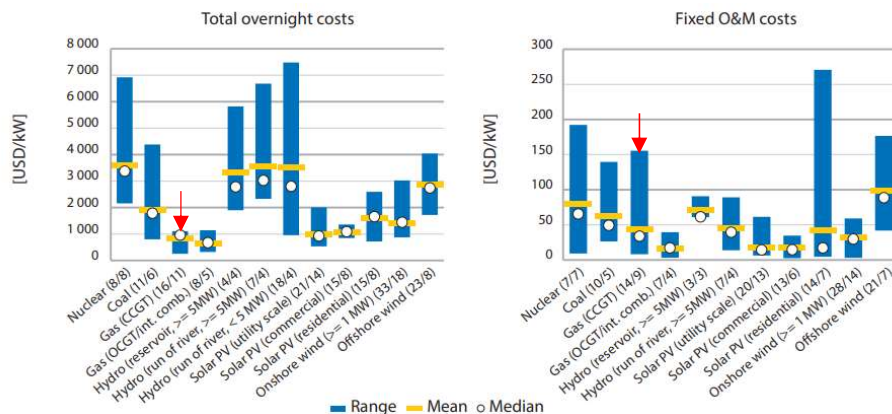


Figure 7: Overnights costs and fixed O&M costs of electricity of different technologies in 2020.

Source: IEA Report, 2020

7.1.14 A horizontal line drawn across the plot illustrates that there are other technologies with LCOE values that are within the same range as the LCOE value of gas (CCGT). These technologies are, utility scale PV (with lower median), hydro-reservoir ≥ 5 MW (with wider range of LCOEs).

7.1.15 According to the EIA 2020 report, the calculated scenarios of LCOE at different discount rates results in a median LCOE of \$70/MWh at 7% discount rate, and increases to \$89/MWh at 10% discount rate. Figure 9 illustrates the impact of discount rate on the LCOE for various technologies including CCGT. From the plot it is evident that the increase in the discount rate from 0 to 20% in the case of CCGT results in a slight increase of the LCOE.

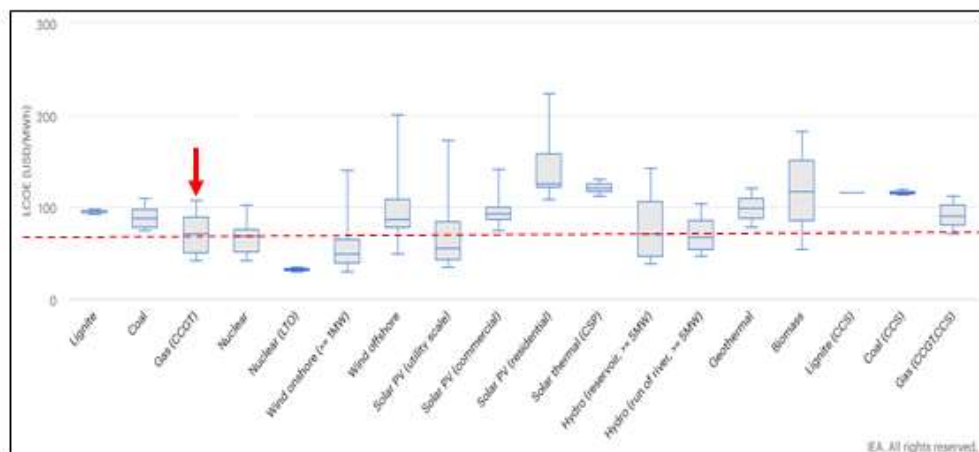


Figure 8: Levelised cost of electricity of different technologies in 2020.

Source: IEA report, 2020

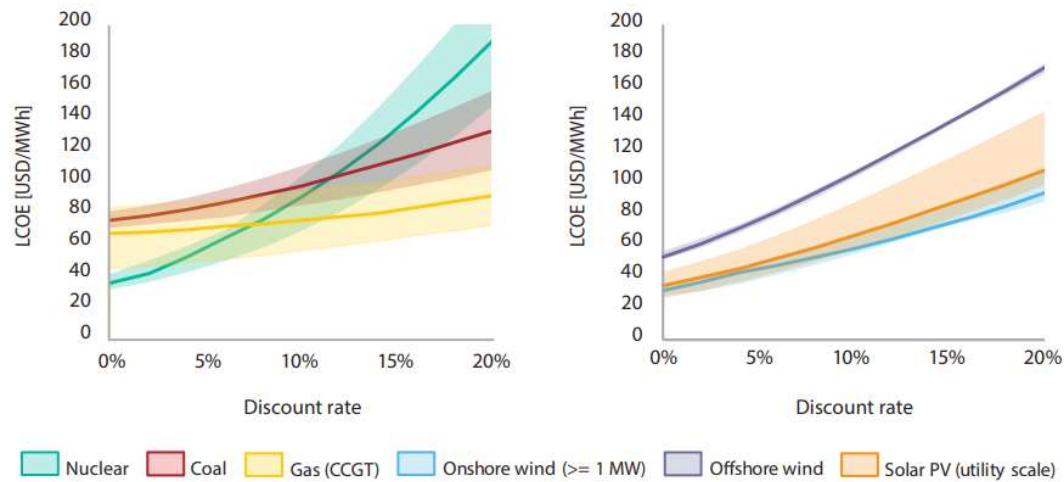


Figure 9: Levelised cost of electricity at various discount rates for various technologies
Source: IEA report 2020

7.1.16 Based on the IEA 2020 international cost report³, Figure 10 shows the LCOE costs at various discount rates from 5% to 20%. A discount rate of 8% would be appropriate for South Africa as it was used in the gazetted IRP2019. As shown in Figure 10, the LCOE for CCGT remains relatively flat at different discount rates. For the assumed discount rate of 8%, the LCOE for CCGT is around US\$60/MWh.

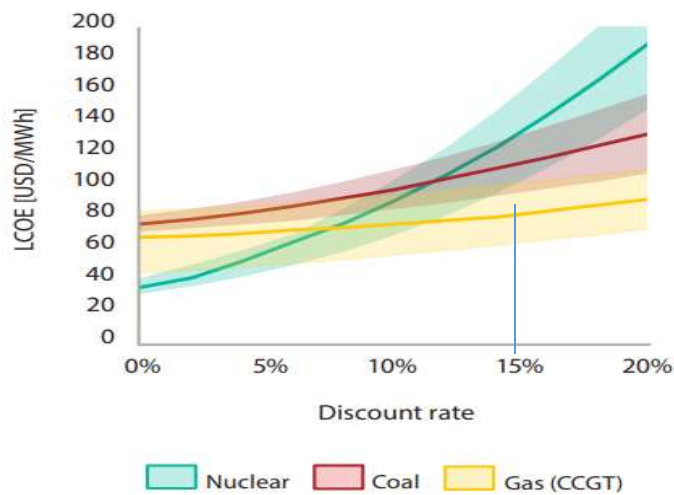


Figure 10: LOCE versus discount rate
Source: IEA report 2020

³ IEA 2020, Projected Cost of Power Generation Report

7.1.17 Using the assumptions in Figure 10 above and a nominal discount rate of 8% and a worst case scenario discount rate of 15%, the LCOE for CCGT technology is expected to range from R1.05 to R1.22/kWh. This range falls below the current blended costs of RE (currently around R2.20/kWh), and it is also lower than prices of coal and nuclear technologies. However, it should also be noted that the prices of BW4 variable Solar PV and Wind power plants are currently around R1.00/kWh, and are expected to be lower in BW5. These variable technologies will however need support from flexible generators and storage options, all of which will come in at an added cost to the average tariff.

NERSA analysis of Fuel Costs

7.1.18 NERSA has noted that Eskom stated in its comments that it will be responsible for procurement of the fuel in terms of a gas supply agreement. It is further stated that Eskom will enter into an agreement with suppliers of the LNG Gas feedstock through an open and fair procurement process.

7.1.19 NERSA also notes concerns of affordability raised by stakeholders given that the gas must be procured in US dollars and that the volatility of the Rand could affect price certainty. This was raised in recognition of the fact that South would largely have to depend on imported gas. It is further argued that this problem would be a complex exercise as it involves hedging on two fronts: hedging against the depreciation of the rand against the US dollar; and hedging against the international gas price volatility. The view of stakeholders therefore is that without significant domestic production of gas, price volatility of gas generated electricity may be inevitable.

7.1.20 The concerns raised by stakeholders on affordability are legitimate and may push the price of electricity generated to excessive levels. As such, NERSA recommends that this risk be shared with the supplier of LNG. The supplier of gas should be tasked with demonstrating how they will mitigate the price volatility of LNG when it responds to bids. NERSA will regulate the gas supplier and will independently ensure that it follows through on the conditions of its bid as well.

7.1.21 As a concluding remark, it is important to note that the above figures of LCOE are indicative cost ranges for CCGT technology. The actual costs will vary widely depending on the region, type of technology chosen, discount rate used, and on delays in completing construction. Furthermore, estimated figures stipulated in Table 1 above are comparable to those reflected above.

8. GENERATOR

8.1 DMRE Generator Proposal

8.1.1 The Section 34 determination states that the electricity must be procured by Eskom and also generated by Eskom.

Stakeholder analysis of the chosen Generator

8.1.2 The determination did not explicitly state who the generator of this new generation capacity would be, stating only that the capacity will be procured by Eskom in paragraph 4. Thus stakeholders did not comment on the chosen generator of this capacity.

NERSA Analysis of Chosen Generator

8.1.3 In light of the fact that this determination is based on the Eskom's application, a due process that emanates from sections 5 and 6 of the New Generation Capacity (published as GNR. 399 in Government Gazette No. 34262 dated 4 May 2011, as amended, was followed in that Eskom was required to submit feasibility studies for the 3 000MW gas project. In response to the application, the minister then issued a Section 34(1) to NERSA to indicate that the new 3 000MW from gas shall be established by Eskom.

8.1.4 It is well understood that once a favourable decision on section 34 determination has been reached, it will enable Eskom to seek a business case approval from relevant committees including other Government departments such as Department of Public Enterprises (DPE) and the National Treasury (NT). A further assessment will be conducted by NERSA during the application for a licence and conclude using financial information that will be submitted at licensing stage to conclude whether Eskom is well positioned to generate this new capacity based on that balance sheet.

9. BUYER

9.1 DMRE buyer proposal

9.1.1 Since Eskom is the developer and generator of the capacity, there is no buyer of this 3000MW capacity from Gas since Eskom is the single buyer of all independently produced generation capacity.

NERSA buyer analysis

9.1.2 According to the Act, the powers to determine the buyer belong to the Minister. The determination of the buyer is therefore a policy issue that is in the hands of the DPE and the DMRE.

9.1.3 However, the Energy Regulator will use its rules and other Regulatory instruments to ensure that only efficiently incurred costs of the project are passed on to end-users.

10. THE PROCURER

10.1 DMRE proposed procurer

10.1.1 The ministerial determination identifies Eskom as the proposed procurer of the new Generation Capacity ("the electricity"). It further states that the electricity shall be procured through one or more tendering procedures which are fair, equitable, transparent, competitive and cost-effective and shall constitute IPP procurement programmes as contemplated in the Regulations.

Stakeholder comment on the chosen procurer

10.1.2 Some stakeholders were of the view that Eskom should not be part of this process as it presents a conflict of interest. They further stated that currently, in the absence of the ISMO, Eskom cannot be chosen as the procurer of this new generation capacity. Eskom itself commented by saying that, whereas Paragraph 4 of the ministerial determination only identifies Eskom as the procurer of the new generation capacity, Eskom is actually the developer and the owner of the new generation facility in question. Eskom also stated in its comments that it will be responsible for procurement of the fuel in terms of a gas supply agreement, and for the transmission connection.

Eskom is therefore of the opinion that paragraph 4 and 6 of the determination does not reflect the true sense of what Eskom applied for to the minister of DMRE.

10.1.3 Furthermore, Eskom stated that it is unclear as to the intent behind the provision determined by the minister in paragraph 6 of the determination attached hereto as **ANNEXURE A**. In Eskom's written request to the Minister for a section 34 determination in respect of the Project, Eskom stated that it did not indicate to the Minister that the Project would be undertaken with the support of other state-owned companies. On the contrary, Eskom is obligated – both by law and by the Draft Determination – to conduct competitive tendering procedures for the selection of its suppliers, including the gas supplier.

10.1.4 Another Stakeholder stated that there is no justification for the procurement of this additional 3000 MW proposed to be done through Eskom as opposed to the standardised procurement entity of the IPP Office for this express purpose of managing the power procurement process with Eskom as the sole off-taker from IPP's. It is interpreted, that this is with the intention of Eskom's exploration of a Public Private Partner (PPP) scheme for their proposed 3000 MW CCGT in Richards Bay, however given the ongoing litigation of this proposed project; consideration should be given to the scenario where the litigation does not fall in Eskom's favour stalling this additional 3 000 MW capacity allocation that could otherwise be mobilised by private developments in the same strategic area of Richards Bay to have capacity added to the grid prior to 2028, provided the complexities in the pricing mechanism are addressed.

NERSA analysis of the chosen procurer

10.1.5 NERSA's analysis of the procurement method included a risk assessment to mitigate perceived risks and whether procurement should be done through Engineer Procure and Construct (EPC) contracts or through the Engineering, Procurement and Construction Management (EPCM) process to insulate the projects from cost overruns and construction delays due to the fact that the capacity will be developed by Eskom. Eskom will have to apply lessons learned from Medupi and Kusile new built plants. NERSA will ensure that costs due to delays and cost overruns are not passed onto consumers of electricity.

The stakeholder that purports that Eskom could be pursuing a PPP scheme in this project may not be accurate because Eskom will enter into an agreement with suppliers of the LNG Gas feedstock through an open and fair procurement process.

11. ECONOMIC ASSESSMENT

11.1 Impact of the procurement of the 3 000MW on the economy

11.1.1 A high-level economic assessment was done to assess the programme's impact on the economy and the Gross Domestic Product (GDP). The study showed that due to the large amount of 3 000 MW that will be developed through this programme, there would be a positive impact on the economy due to the fact that less OCGT peaking generators will be utilised by Eskom and that the programme will reduce load-shedding.

11.1.2 Further, positive spin-offs of this programme are likely to be realised in the piped gas industry also regulated by NERSA. The project has the potential to cater for additional existing gas demand beyond the Eskom's CCGT power generation plant. The gas industry in South Africa is currently facing gas supply threats due to the looming declining gas production in the Pande/Temane gas fields in Mozambique expected from 2025 as well as the production constraints of methane rich gas in the Sasol plant in Secunda. South Africa currently imports about 160 million gigajoules per annum of natural gas from the Pande/Temane gas fields that is mostly used in the country's industrial sector by Sasol for its chemicals and petrochemical operations in Secunda and Sasolburg as well as external customers in various industries including - Iron & steel, non-ferrous metals, pulp & paper, food and tobacco, and non-metallic minerals.

11.1.3 According to DMRE (2021), the industrial sector consumed about 51% of the total energy supplied in South Africa in 2018, with gas meeting about 29% of that energy demand. The chemical sub-sector is the biggest consumer of gas, followed by iron & steel, pulp & paper and food and tobacco.⁴

11.1.4 The users of gas, as a combined total, generate over R150 billion in revenue a year and employ over 50 000 people. Therefore, the looming gas supply threat could have detrimental effect to the economy. As such this LNG-to-power

⁴ Department of Mineral Resources and Energy (DMRE), 2021. The South African Energy Sector Report 2021

programme in Richards Bay presents the opportunity for the country to avert this negative impact. The supporting LNG infrastructure to be developed to ensure reliable gas fuel supply to the Eskom's CCGT plant can be designed such that additional capacity is made available to supply markets beyond the power generation plant.

11.1.5 In particular, the LNG receiving terminal can be designed to ensure that there is third party access to the capacity not committed to the CCGT power plant. This would require the LNG terminal owner and operator to provide for excess LNG storage capacity beyond the power plant demand that may be used to store gas for regasification and supply to the brownfields markets along the Richards Bay and Durban corridor via the existing Lilly Pipeline and Sasol's distribution and transmission gas supply nodes. Further, LNG can be used to facilitate the development of a small-scale LNG market around the Richards Bay area. This also provide the opportunity to create competition in the market as new entrants (outside the LNG terminal Operator) may provide the small-scale LNG services to make gas available beyond the power generation plant.

11.1.6 The programme will therefore have a positive overall impact on the economy of South Africa and could result in improved GDP. The increase in the GDP could result in many positive possibilities including the creation of jobs and improved socio-economic status.

12. BENCHMARKING

12.1 Benchmark studies

12.1.1 A benchmark study was performed that investigated global trends while considering the fact that South Africa needs South African solutions that will create an enabling environment for the Electricity Supply Industry (ESI) and ensure that electricity prices remain affordable to end-users.

12.1.2 The benchmark study showed that the program is in line with what other countries do to complement renewables, such as Germany, Italy, Ireland and many others OECD countries which use gas to counter the variability of renewables, leading to a more balanced system. Figure 11 shows typical examples of the complementary relationship between renewable energy and gas recently experienced in Brazil and California.

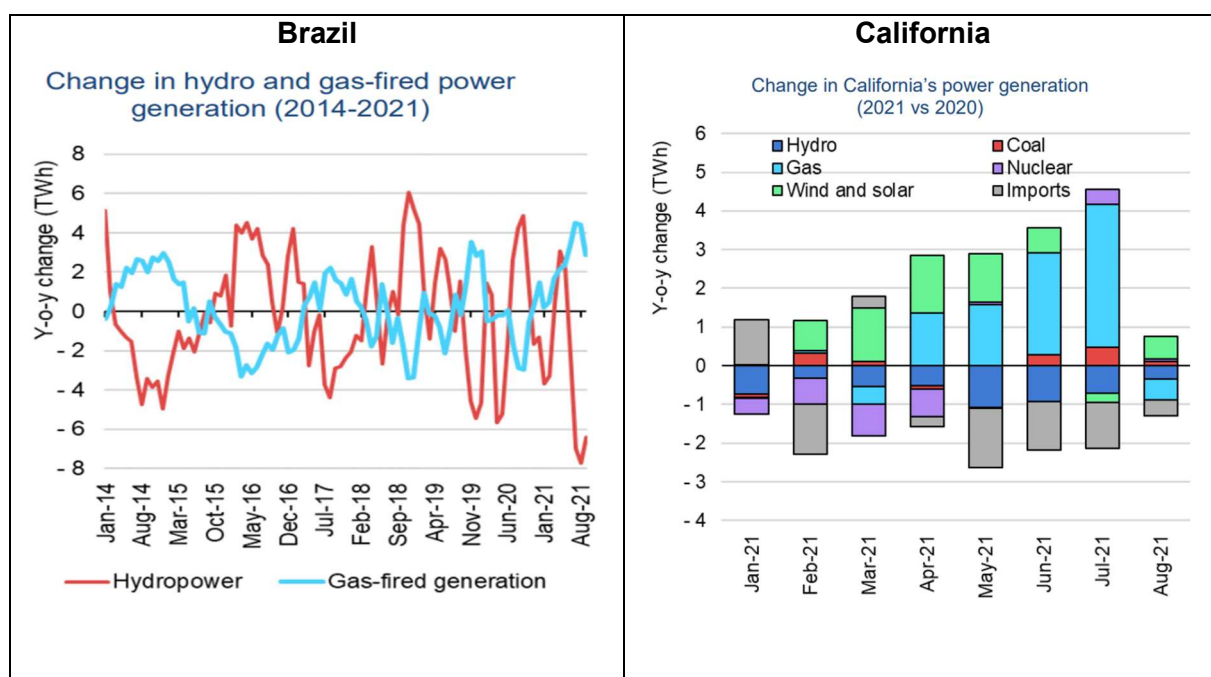


Figure 11: Gas and renewables interplay in Brazil and California

Source: IEA (2021). Gas Market Report Q4: 2021

12.1.3 According to reports from the IEA (2021) and International Gas Union (2022), Brazil (a hydro-rich country where hydroelectric power normally makes up about 64% of the total generation capacity) has experienced severe historic drought in 2021 that caused hydro reservoirs to drop to 17% below the normal 5-year average thus resulting in low hydropower output and forcing the country to import more LNG to narrow the power supply gap. In 2021, Brazil imported

about 7.5 Mtpa of LNG mainly for power generation over and above the normal gas pipeline imports used for other various applications. The reliance on gas-fired power decreases when the hydro-electric power output increases thus allowing for the related gas volumes to be released for other uses in the domestic market or international trade.

12.1.4 California is also known for its renewable energy-rich power generation mix—dominated by hydro, solar and wind. On average, solar and wind energy contributes above 25% of the power generation capacity while hydro alone is accounted for 15% over the last decade. Severe drought experienced in the summer of 2021 in the region led to a steep decline in hydropower output and increased use of gas-fired power to fill the power supply gap. The use of thermal generation, particularly gas-fired power has become a common phenomenon in most jurisdictions due to variable weather and hydrology conditions that affect the availability of the wind, solar and hydro renewable resources.⁵⁶

12.1.5 It should be noted that the battery energy storage solution is viewed as a natural complement to variable renewable energy from the environmental standpoint than gas-fired power. While the renewable energy and battery storage combination may be a preferred sustainable solution to clean energy, the availability and performance of the battery energy storage system at grid-scale puts the efficiency of this combination into question. Battery storage is arguably efficient for meeting short-term energy demand that cannot be served by renewables but cannot be easily relied on to provide a long-term energy storage solution to service high seasonal demands which are often difficult to predict accurately as they are influenced by unstable weather patterns for prolonged periods. Further, BESS are capable of self-discharging even when not in use thus reducing the amount of energy available for later use by the grid or customer when renewable energy supply is low or unavailable. This further justifies the need for gas in the generation mix as a flexible and reliable technology to balance the energy system irrespective of the varying demands.

12.1.6 The reliability of gas supply becomes a critical component to the functioning of the CCGT power plant due to high capacity factors. As seen recently, Germany had to recommission its coal plants when it lost the supply of gas from Russia

⁵ IEA, 2021. Gas Market Report Q4: 2021

⁶ International Gas Union (IGU). Global Gas Report 2022

and could not balance supply and demand as a result. This risk must be mitigated in South Africa by diversifying its gas supply portfolio and source gas from different countries to manage gas fuel supply risks to the power plant in the absence of indigenous gas sources. The use of gas storage systems as a contingency plan to keep sufficient quantities of gas is another complementary solution used by mature markets in Europe and Asia-Pacific to avert the gas supply risks. These countries buy and store sufficient quantities of gas during low demand season (particularly when the price is lower) for use when supply is constrained due to geopolitical or other reasons.

13. CONCLUSIONS

13.1 NERSA conclusion on the determination

13.1.1 The benefits of this 3 000MW from gas power technology in a form of CCGT are that it will provide the system with much needed reliable power during peak periods and reduce the over-usage of OCGTs.

13.1.2 Another benefit of gas power technology in the form of CCGT is to strike the correct balance in the energy mix as there is a fundamental change towards renewable resources. Therefore, a mid-merit Combined Cycle Power Plant is a technically feasible solution that is required to supplement batteries and close the impending base-load gap.

13.1.3 Furthermore, positive spin-offs would be realised in the piped gas industry. The LNG receiving terminal to be developed to ensure reliable gas fuel supply to the CCGT power plant would create opportunities for South Africa to diversify its gas supply sources and buy gas from different countries in the international market. Further, this will reduce the country's overdependence on Mozambique gas and avert the gas supply risks faced by the existing market relying from the depleting Pande/Temane gas resources. Further opportunities exist to establish a new small-scale LNG market around the Richards Bay/ Durban Corridor, thus lessening the dependence on electricity from the grid.

13.1.4 Although there are a number of benefits with establishing this capacity, NERSA does not concur to this determination due to the fact that the Section 34 determination as presented, falls outside the scope of what NERSA is required to concur with in Section 34 of the Act.

Recommendation

13.1.5 NERSA therefore does not concur with the Ministerial determination for the procurement of 3 000 MW from Gas technology and recommends that the project should not be brought back to the regulator under the Section 34 determination process again, but rather that Eskom should go back to the minister and request a diversion from the IRP 2019 as provisioned in the Act under section 10(2)g, which states that:

(g) evidence of compliance with any integrated resource plan applicable at that point in time or provide reasons for any deviation for the approval of the Minister; and
(h) such other particulars as the Minister may prescribe.

13.1.6 Since this capacity is a diversion from the IRP 2019, NERSA recommends that the project be licensed as per section 10(2)g above. Furthermore, since the purpose of the section 34 determination process premises on giving a go ahead to the commencement of procurement processes, and; whereas this project is already advanced in its development, bears testimony to the fact that the concurrence process is redundant in this case in our view.

ANNEXURE A: THE MINISTERIAL DETERMINATION ON THE PROCUREMENT OF 3 000 MW NEW GENERATION CAPACITY FROM GAS

GOVERNMENT NOTICE

DEPARTMENT OF MINERAL RESOURCES AND ENERGY

No. 2022

DETERMINATION UNDER SECTION 34(1) OF THE ELECTRICITY REGULATION ACT, 2006 (ACT NO. 4 OF 2006)

The Minister of Mineral Resources and Energy ("the Minister"), in consultation with the National Energy Regulator of South Africa ("NERSA"), acting under section 34(1) of the Electricity Regulation Act, 2006 (Act No. 4 of 2006) (as amended) (the **ERA**) and the Electricity Regulations on New Generation Capacity (published as GNR. 399 in Government Gazette No. 34262 dated 04 May 2011, as amended) ("Regulations"), has determined as follows:

1. that 3000 MW of gas power new generation capacity at Richards Bay is needed to be procured to contribute towards energy security.
2. the electricity procured shall target connection to the Grid as soon as reasonably possible and not beyond the year 2028.
3. there must be a prior agreement by NERSA on the cost recovery mechanism.
4. the generation capacity must be procured by Eskom Holdings SOC Ltd through one or more tendering procedures which are fair, equitable, transparent, competitive and cost effective.
5. procurement of primary energy and its associated logistics infrastructure must be done in a manner supportive of State's participation through other State Owned Companies which have the relevant mandate(s)

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ANNEXURE B: SUMMARY OF STAKEHOLDER COMMENTS (GAS TECHNOLOGY)

The following stakeholders commented on the proposed ministerial determination for 3000MW CCGT gas plant to be developed and owned by Eskom SOC Limited at Richards Bay. Further below is the detailed stakeholder responses and NERSA's comments.

1. Business Unity South Africa (BUSA)
2. Centre for Environmental Rights (CER), Natural Justice (NJ), Groundwork and South Durban Community Environmental Alliance (SDCEA)
3. Eskom
4. Energy Intensive Users Group (EIUG)
5. Gigajoule International
6. Meridian Economics
7. Minerals Council South Africa
8. Moondream
9. Phakwe Group
10. Richards Bay Industrial Development Zone (RBIDZ)
11. SASOL
12. Umhlathuze City Development

1. Business Unity South Africa (BUSA)

No.	Questions from the consultation paper	Stakeholder Comment	NERSA ANALYSIS ON STKEHOLDER COMMENT
i. TECHNOLOGY			
1.	In your view, is gas the type of technology that is in line with best practice and will the determined capacity ensure security of supply?	Yes. Whilst battery storage would allow for some grid stability; it will be insufficient as SA winds down on its reliance on coal. Gas is an essential transition fuel. Furthermore, an offtake in the electricity sector would be a solid underpinning for investment that would similarly support the industrial use of gas. Gas is a proven, reliable, and flexible technology that can be used for peaking, mid-merit or baseload power generation. Natural gas also has far fewer emissions than coal (about 50% less when used for power generation) and is the lowest fossil fuel emitter. If the supply of gas/LNG is secured, the supply of electricity will be secured.	• The 3 000MW to be generated from Gas technology is in line with best practice because it will improve the supply-demand imbalance in the country and support the Electricity Regulations Act's objective to promote the use of diverse energy sources and energy efficiency. • It will also support the Government's initiatives to bring affordable, reliable and cleaner energy to power consumers.
2.	Provide your comments on the availability of natural gas in South Africa	The availability of gas can be secured as LNG or from exploration and production projects (upstream). LNG is largely available and gives access to multiple sources of supply worldwide through the international LNG markets. For this purpose, a gas power plant should be connected to an LNG import terminal via pipeline.	• The gas availability in South Africa is a concern. However, Eskom is in a process of engaging with partners like Transnet whose role will be to participate in Liquefied Natural Gas (LNG) receiving infrastructure, regasification, gas transmission and distribution.
3.	Is the determined capacity sufficient to ensure	The EAF of Eskom's coal power generation fleet is deteriorating faster than expected. Sufficient gas to power capacity would depend on i) decommissioning of existing	• The electricity system has been constrained for most of the past 14

	uninterrupted supply of electricity?	coal capacity and ii) the battery storage capacity that would accompany renewables. Based on the current IRP, we believe that more of gas capacity will be required. This consultation paper would appear to address procurement of 3GW from IPPs using gas in terms of Eskom's 3GW gas to power plans in Richards Bay. In accordance with the IRP, a further 3GW of gas to power is required, rendering this capacity as per the NERSA consultation document, insufficient in the context of Eskom's deteriorating coal fleet efficiencies and needs to convert coal generation plants to gas. The overall additional capacity needs to be looked at compared to the overall demand, including the type of generation (peak, mid merit, baseload). If Eskom indicated a need for 4000 to 6000 MW additional baseload generation capacity and this had not been addressed, then the supply will not be enough for the demand. However it depends on the utilisation of the gas.	years. There is also a fundamental change in the energy mix towards renewable resources. Therefore, balanced mid-merit Combined Cycle Power Plant is a technically feasible solution to help with the power crisis.
4.	Provide what you consider to be the risks and challenges associated with this new capacity.	• Not securing enough funding as gas is a fossil fuel • The terms of the PPA – such that it would afford investors an appropriate return over a mutually agreed term – and the consequential impact on tariffs • The dispatch schedule and how this could impact the financial viability of renewables • The LNG import terminal in Richards Bay is still in a very early stage (RFI), and the outlook of the first gas to be imported is very unclear. • Eskom's ESIA for the 3 GW CCGT is being challenged in front of courts. • More globally, the main challenges are: - progress/development of LNG receiving facility (FSRU/onshore) to match the deadline of the 3GW required	• The risk of the availability of gas can be minimized if Eskom can find local partners to supply reliable gas, at relatively stable price to avoid volatility of gas price. Hence, domestic partners can curb international gas volatility. • Also, gas market aggregation can mitigate this risk whereby a lot of players can compete or use bargaining power to buy it cheaper.

5.	Provide your comments on whether the country will benefit more optimally from gas and the benefits of establishing a gas market.	In addition to gas to power needs, there is a lot of unsatisfied gas demand in the country, especially for industrials using gas in their facility processes. This would clearly support existing businesses to grow and also current coal-user to switch to gas which will have a pretty positive impact on the decarbonisation journey engaged by most of the industry players. Supplying gas to small industrial users can also help them go off the grid and release some pressure on the country grid. So, making gas available would definitely have a positive effect on the economy.	• A gas market with domestic partners will curb gas-price volatility caused by the international markets. Therefore, an established gas market in South Africa will ensure accessibility and affordability.
6.	What do you think is the optimal timelines within which this capacity should be made available on the grid, considering the urgency that is required to deal with load shedding in the country?	Considering the power deficit in South Africa and the urgency to speed up power generation capacity, focus should be given the delivery of the “fastest” gas to power projects. Given the opportunity for the utilisation of existing infrastructure, it follows that gas to power off the Rompco pipeline should initially be prioritised, with Richards Bay, Coega, Mossel Bay and Saldanha to follow in parallel.	• The optimal timelines are as emphasized in the determination which must not be beyond 2028. • In its feasibility study, Eskom indicated indicative timelines of commissioning different units of which all of the 4 units will be commissioned by no later than the end of year 2028.
ii. COST RECOVERY MECHANISM AND THE IMPACT ON TARIFFS			
1.	Should NERSA continue with the current cost recovery mechanism stipulated in section 13 of the current Multi-Year Price Determination Methodology (MYPD) and section 14 of NERSA's CRM Rules? If not, please propose the most suitable cost recovery mechanism	• PPA should provide for a fixed fee/tariff for available energy and an additional fee for energy dispatched over a lengthy period [15 years+], adjusted for inflation • Many of the risks would therefore be transferred to the IPP, albeit that provision may need to be made for gas price shocks • This cost would then be passed on to the consumer as per the existing MYPD but would be predictable based on the PPA	• Cost Recovery Mechanism should have not been included as part of section 34 determination as it falls within the scope of the MYPD4 Methodology and CRM Rules.

	for the gas project that would ensure that investment in a gas-to-power facility is economic, resulting in a good opportunity cost for the investor over the life of the asset, while also ensuring that electricity remains affordable.	The cost recovery mechanism seems suitable and even necessary to enable LNG import. However, the MGP mechanism does have some detrimental effects (commodity risk for LNG importers); therefore for tendering process and in order to get the most competitive offer, no MGP should be applied.	
2.	Provide your thoughts on the costs that will be associated with establishing the new allocated generation capacity in line with the mandate to ensure the long-term sustainability of the electricity supply industry, as well as affordability and equity.	Considering a similar project cost for the power generation unit (a given capacity), the difference can be made in the cost linked with the gas supply chain, and especially the gas import facilities. Indeed, investment for an FSRU project could be about 500M US\$ and an onshore LNG terminal could be around 1.2B US\$.	EPRI and Lazard provide the overnight cost of 3 000MW produced from gas. Although these costs are indicative but they provide an indication how much it will cost to produce 3 000MW from gas.
3.	Provide your thoughts on the impact of adding the additional capacity of 3000MW from gas in relation to NERSA's mandate to ensure the long term sustainability of the electricity supply industry as well as affordability and equity?	- Security of supply: Whilst battery storage would allow for some grid stability, it will be insufficient as SA winds down on its reliance on coal. Gas would be an essential transition fuel - Affordability: The market should be tested in a competitive bidding process - Equity: Underpinned by the terms of the PPA whereby IPP will earn income on all available capacity, whether or not it is dispatched	Additional megawatts will ensure the long term sustainability of the electricity supply industry. However, procuring 3 000MW from gas must be done within the correct legislative requirements as it does not fall within a section 34 determination.
3.	How should the cost associated with the fuel for this technology be	Preferable to pass the majority of the risk to the IPP, but a level of adjustment would be required for price shocks (up or down)	The CRA or any other regulatory price methodology that will be available at

	dealt with within the cost recovery mechanism in order to mitigate against rapid changes in prices of gas?	• To the extent that this cannot be passed on to the consumer, the fiscus will need to subsidise • It has to be noted that in the case of gas supplied from exploration and production / upstream development, the price of gas wouldn't have necessary to be directly linked with international LNG prices.	that time will be used to mitigate against rapid changes in prices of gas.
4.	What will be the anticipated tariff impact to customers or end users in R/kWh if 3 000MW of gas-to-power is procured?	• Full details of this would only be available on a completed RFP process. • The outcome of the recently announced RMIPPPP gas to power projects is not a reasonable proxy as 1) the technologies and basis for operations are significantly different and 2) IPPP tender process that required JKM to be used as fuel cost proxy following on NERSA's maximum price methodology. • The RFP process should not be prescriptive on LNG/gas pricing to ensure a competitive bidding outcome at lowest possible gas pricing based on alternative indexes and market pricing.	The revenue requirement or any other price determination methodology that will be available at that time, will be used to calculate the tariff impact of 3 000MW.
5.	Provide your comments on the socio-economic impact of the determined capacity in terms of the number of jobs that can be created, including the objective of a just transition from coal to clean technologies.	As for the socio-economic benefits provided to currently unsatisfied gas demand, any gas supply project will have a positive impact. The geographic area reached by this gas will depend on the transport costs to reach this demand. Projects that use existing infrastructures, such as the Rompco pipeline and/or the Moss gas offshore infrastructure, will reach a larger area at a lower cost due to the use of existing infrastructure.	• Since the Electricity Supply Industry (ESI) is transitioning in line with Just Energy Transition, the 3 000MW from gas technology will minimize the impact of Eskom staff members who might lose their jobs because of some of the coal fired power stations reaching the end of their useful lives.

			• It will also create jobs to the locals during different stages of project life cycle. •
iii. ENVIRONMENTAL IMPACT OF THE NEW GENERATION CAPACITY OF 3 000 MW FROM GAS AT RICHARDS BAY			
1.	Do you think that it is proper for the Energy Regulator to deviate from its mandate and consider other Government Departments mandates when concluding section 34 determinations?	No, but NERSA should consider all aspects from an SA perspective, including: • Security of supply • Electricity pricing • Just Transition • The environment	• This determination derives its mandate from the Electricity Regulations Act. NERSA does not have powers over mandates of other Government Departments.
2.	Is it proper for the Energy Regulator to consider matters that will be dealt with at licensing stage during the concurrence process?	Yes. It will be a futile exercise to proceed with this concurrence if it is eventually blocked for environmental reasons.	• This determination derives its mandate from the Electricity Regulations Act. NERSA does not have powers over mandates of other Government Departments.
IV. PROCUREMENT PROCESS UNDER THE IPP PROCUREMENT PROGRAMME			
1.	Provide your thoughts on Eskom as the chosen procurer of this new generation capacity.	• Appropriate in a world where Eskom is separated • Not appropriate for Eskom Generation to have visibility on the PPA, as it is conflicted	• Another process should be followed in this regard as this is not a section 34 determination process
2.	Provide your thoughts on the method chosen for the procurement of the new generation capacity.	Seems fair provided that Eskom Generation has no visibility. Refer comments above on NERSA and gas energy pricing basis under the RMIPPPP.	• Another process should be observed in this regard as this is not a section 34 determination process

3.	Section 5 of the Section 34 determination states that the procurement of primary energy and its associated logistic infrastructure must be done in a manner supportive of the State's participation through other State Owned Companies that have the relevant mandate(s). Provide your thoughts on this requirement.	The state participates by virtue of its ownership of the transmission and distribution infrastructure	This falls outside of the Regulator's mandate. However, Public (SOEs)-Private-Partnerships (PPPs) are known for producing most desirable outcomes in building mega projects.
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2. CER, NJ & SDCEA

No.	Questions from the consultation paper	Stakeholder Comment	NERSA ANALYSIS ON STKEHOLDER COMMENT
i. TECHNOLOGY			
7.	In your view, is gas the type of technology that is in line with best practice and will the determined capacity ensure security of supply?	No, because wind and solar, and/or storage are replacing gas and coal plants that provide smaller-scale power, relieve the energy grid during peak hours, and help facilitate the further integration of renewable energy into the grid.	- The 3 000MW to be generated from Gas technology is in line with best practice because it will improve the supply-demand imbalance in the country and support the Electricity Regulations Act's objective to promote the use of diverse energy sources and energy efficiency. - It will also support the Government's initiatives to bring affordable, reliable

			and cleaner energy to power consumers. • However, this is not a section 34 determination. Therefore, another process to facilitate this application should be followed
8.	Provide your comments on the availability of natural gas in South Africa	• The stakeholder referenced documents of which it did not provide as part its submission/comments	• The gas availability in South Africa is a concern. However, Eskom is in a process of engaging with partners like Transnet whose role will be to participate in Liquefied Natural Gas (LNG) receiving infrastructure, regasification, gas transmission and distribution. • However, this analysis provided above does not take away the fact that this is not a section 34 determination. Therefore, another process to facilitate this application should be followed
9.	Is the determined capacity sufficient to ensure uninterrupted supply of electricity?	• The determined capacity from gas is not needed to ensure uninterrupted supply of electricity. Our response to this question is that renewables, with other flexible generation, can meet the demand currently being proposed for gas.	• The electricity system has been constraint for most of the past 14 years. There is also a fundamental change in the energy mix towards renewable resources. Therefore, balanced mid-merit Combined Cycle

			Power Plant is a technically feasibly solution to help with power crisis. • However, this analysis provided above does not take away the fact that this is not a section 34 determination. Therefore, another process to facilitate this application should be followed
10.	Provide what you consider to be the risks and challenges associated with this new capacity.	• The risks are that these type of projects become stranded assets and/or unable to operate for their full expected lifespans due to carbon constraints, climate risks and other impacts.	• The risk of the availability of gas can be minimized if Eskom can find local partners to supply reliable gas, at a required pressure and a relatively stable price to avoid volatility of gas price. Hence, domestic partners can curb international gas volatility. • Also, gas market aggregation can mitigate this risk whereby a lot of players can compete or use bargaining power to buy it cheaper. • However, this analysis provided above does not take away the fact that this is not a section 34 determination. Therefore, another process to facilitate this application should be followed
11.	Provide your comments on whether the country will	• Gas power would create significant harms and risks to South Africa. At a time when urgent action is necessary	• A gas market with domestic partners will curb gas-price volatility caused by

	benefit more optimally from gas and the benefits of establishing a gas market.	to avert a climate disaster, and when alternative, cost-effective, and cleaner renewable and storage technologies are available, South Africa should not promote a gas market.	the international markets. Therefore, an established gas market in South Africa will ensure accessibility and affordability. • However, this analysis provided above does not take away the fact that this is not a section 34 determination. Therefore, another process to facilitate this application should be followed
12.	What do you think is the optimal timelines within which this capacity should be made available on the grid, considering the urgency that is required to deal with load shedding in the country?	• We have already stated above that renewables and storage are more appropriate and better suited to closing the energy gap than gas. Further, in line with the cautious approach dictated by NEMA and the need for regular review of electricity demand and cost, as necessitated by the IRP 2019, it is not appropriate that this determination has been published outside of the IRP 2019 and in the absence of any key modelling and impact assessments.	• The optimal timelines are as emphasized in the determination which must not be beyond 2028. • In its feasibility study, Eskom indicated indicative timelines of commissioning different units of which all of the 4 units will be commissioned by no later than the end of year 2028. • However, this analysis provided above does not take away the fact that this is not a section 34 determination. Therefore, another process to facilitate this application should be followed

ii. COST RECOVERY MECHANISM AND THE IMPACT ON TARIFFS			
1.	Should NERSA continue with the current cost recovery mechanism stipulated in section 13 of the current Multi-Year Price Determination Methodology (MYPD) and section 14 of NERSA's CRM Rules? If not, please propose the most suitable cost recovery mechanism for the gas project that would ensure that investment in a gas-to-power facility is economic, resulting in a good opportunity cost for the investor over the life of the asset, while also ensuring that electricity remains affordable.	- The volatile price of gas makes the proposed gas determination both risky for the investor (i.e. if gas prices are low) and for rate-payers if gas prices are high as they are now. - The risk of a facility becoming a stranded asset long before the end of the typical 40-year lifespan of a new gas plant is a particular risk.	Cost Recovery Mechanism is valid. However, it should not be form part of section 34 determination as it falls within the scope of the MYPD4 Methodology and CRM Rules.
2.	Provide your thoughts on the costs that will be associated with establishing the new allocated generation capacity in line with the mandate to ensure the longterm sustainability of the electricity supply industry, as well as affordability and equity.	A recent analysis of the costs of developing the 3000 MW combined-cycle power plant facility at Richards Bay calculated that it would cost 47 billion ZAR to build the plant and the associated floating storage and regasification unit at Richards Bay to supply the plant with gas.	- EPRI and Lazard provide the overnight cost of 3 000MW produced from gas. Although these costs are indicative but they provide an indication how much it will cost to produce 3 000MW from gas. - However, this analysis provided above does not take away the fact that this is not a section 34 determination. Therefore, another process to

			facilitate this application should be followed
3.	How should the cost associated with the fuel for this technology be dealt with within the cost recovery mechanism in order to mitigate against rapid changes in prices of gas?	• Long term contracts and provision of government subsidies (to maintain a given gas price in the face of global spikes) both result in many problems and distortions. Fossil fuel subsidies are simply unjustifiable today.	• The CRA or any other regulatory price methodology that will be available at that time will be used to mitigate against rapid changes in prices of gas. • However, this analysis provided above does not take away the fact that this is not a section 34 determination. Therefore, another process to facilitate this application should be followed
4.	What will be the anticipated tariff impact to customers or end users in R/kWh if 3 000MW of gas-to-power is procured?	• The exact numbers will depend on the capacity factor of the plant and on gas prices. However, an analysis of the tariffs that the Karpowerships would have generated in October 2021 were about ZAR 6/kWh – or over ten times that of renewables in the same period.⁸⁰ These are exceptionally high costs, which cannot be justified.	• The revenue requirement or any other price determination methodology that will be available at that time, will be used to calculate the tariff impact of 3 000MW. • However, this analysis provided above does not take away the fact that this is not a section 34 determination. Therefore, another process to facilitate this application should be followed
5.	Provide your comments on the socio-economic impact of the determined capacity in terms of the	• While short-term jobs will be created during the plant construction phase, gas plant operations require	• Since the Electricity Supply Industry (ESI) is transitioning in line with Just Energy Transition, the 3 000MW from gas technology will minimize the

	number of jobs that can be created, including the objective of a just transition from coal to clean technologies.	notoriously few employees – perhaps a few dozen full-time employees will be needed to run the plant.	impact of Eskom staff members who might lose their jobs because of some of the coal fired power stations reaching the end of their useful lives. • It will also create jobs to the locals during different stages of project life cycle. • However, this analysis provided above does not take away the fact that this is not a section 34 determination. Therefore, another process to facilitate this application should be followed
iii. ENVIRONMENTAL IMPACT OF THE NEW GENERATION CAPACITY OF 3 000 MW FROM GAS AT RICHARDS BAY			
1.	Do you think that it is proper for the Energy Regulator to deviate from its mandate and consider other Government Departments mandates when concluding section 34 determinations?	• NERSA should not, and cannot, stray from its mandate - which lies in enacted legislation, the NERSA Act and ERA.	• This determination derives its mandate from the Electricity Regulations Act. NERSA does not have powers over mandates of other Government Departments.
2.	Is it proper for the Energy Regulator to consider matters that will be dealt with at licensing stage during the concurrence process?	• NERSA's legal obligations under the ERA and NERSA Act, including the Constitution, apply at all stages of decision-making - they cannot be deferred or kicked down the line for consideration at later stages including licensing.	• This determination derives its mandate from the Electricity Regulations Act. NERSA does not have powers over mandates of other Government Departments.
IV. PROCUREMENT PROCESS UNDER THE IPP PROCUREMENT PROGRAMME			

1.	Provide your thoughts on Eskom as the chosen procurer of this new generation capacity.	• Eskom should not be forced to lock itself into unaffordable PPAs using expensive gas to power technologies, which are also at risk of becoming stranded assets, and contributing to high electricity costs	• The chosen Buyer falls outside of the NERSA mandate.
2.	Provide your thoughts on the method chosen for the procurement of the new generation capacity.	• The draft determination states that the electricity procured from the new generation capacity will be procured through tendering processes which are “<i>fair, equitable, transparent, competitive and cost effective</i>”. We dispute that previous tender processes have been fair, equitable, cost-effective and particularly, we dispute that they have been transparent.	• Procurement process falls outside of the NERSA mandate. These powers are for other Government Departments to execute.
3.	Section 5 of the Section 34 determination states that the procurement of primary energy and its associated logistic infrastructure must be done in a manner supportive of the State’s participation through other State Owned Companies that have the relevant mandate(s). Provide your thoughts on this requirement.	• We do not have specific comments on this requirement, suffice to say that we object to the draft gas determination as a whole.	• This falls outside of the Regulator’s mandate. However, Public (SOEs)-Private-Partnerships (PPPs) are known for producing most desirable outcomes in building mega projects.

3. Eskom

No.	Questions from the consultation paper	Stakeholder Comment	NERSA ANALYSIS ON STKEHOLDER COMMENT
i. TECHNOLOGY			
13.	In your view, is gas the type of technology that is in line with best practice and will the determined capacity ensure security of supply?	• The 3 000MW to be generated from Gas technology is in line with best practice because it will improve the supply-demand imbalance in the country and support the Electricity Regulations Act's objective to promote the use of diverse energy sources and energy efficiency. • It will also support the Government's initiatives to bring affordable, reliable and cleaner energy to power consumers.	
14.	Provide your comments on the availability of natural gas in South Africa	• The gas availability in South Africa is a concern. However, Eskom is in a process of engaging with partners like Transnet whose role will be to participate in Liquefied Natural Gas (LNG) receiving infrastructure, regasification, gas transmission and distribution.	
15.	Is the determined capacity sufficient to ensure uninterrupted supply of electricity?	• The electricity system has been constraint for most of the past 14 years. There is also a fundamental change in the energy mix towards renewable resources. Therefore, balanced mid-merit Combined Cycle Power Plant is a technically feasibly solution to help with power crisis.	
16.	Provide what you consider to be the risks and challenges associated with this new capacity.	• The risk of the availability of gas can be minimized if Eskom can find local partners to supply reliable gas, at a required pressure and a relatively stable price to avoid	

		volatility of gas price. Hence, domestic partners can curb international gas volatility. • Also, gas market aggregation can mitigate this risk whereby a lot of players can compete or use bargaining power to buy it cheaper.	
17.	Provide your comments on whether the country will benefit more optimally from gas and the benefits of establishing a gas market.	• A gas market with domestic partners will curb gas-price volatility caused by the international markets. Therefore, an established gas market in South Africa will ensure accessibility and affordability.	
18.	What do you think is the optimal timelines within which this capacity should be made available on the grid, considering the urgency that is required to deal with load shedding in the country?	• The optimal timelines are as emphasized in the determination which must not be beyond 2028. • In its feasibility study, Eskom indicated indicative timelines of commissioning different units of which all of the 4 units will be commissioned by no later than the end of year 2028.	
ii. COST RECOVERY MECHANISM AND THE IMPACT ON TARIFFS			
1.	Should NERSA continue with the current cost recovery mechanism stipulated in section 13 of the current Multi-Year Price Determination Methodology (MYPD) and section 14 of NERSA's CRM Rules? If not, please propose the most suitable cost recovery mechanism for the gas project that	No response provided	

	would ensure that investment in a gas-to-power facility is economic, resulting in a good opportunity cost for the investor over the life of the asset, while also ensuring that electricity remains affordable.		
2.	Provide your thoughts on the costs that will be associated with establishing the new allocated generation capacity in line with the mandate to ensure the longterm sustainability of the electricity supply industry, as well as affordability and equity.	• EPRI and Lazard provide the overnight cost of 3 000MW produced from gas. Although these costs are indicative but they provide an indication how much it will cost to produce 3 000MW from gas.	
3.	How should the cost associated with the fuel for this technology be dealt with within the cost recovery mechanism in order to mitigate against rapid changes in prices of gas?	• The CRA or any other regulatory price methodology that will be available at that time will be used to mitigate against rapid changes in prices of gas.	
4.	What will be the anticipated tariff impact to customers or end users in R/kWh if 3 000MW of gas-to-power is procured?	• The revenue requirement or any other price determination methodology that will be available at that time, will be used to calculate the tariff impact of 3 000MW.	

5.	Provide your comments on the socio-economic impact of the determined capacity in terms of the number of jobs that can be created, including the objective of a just transition from coal to clean technologies.	Since the Electricity Supply Industry (ESI) is transitioning in line with Just Energy Transition, the 3 000MW from gas technology will minimize the impact of Eskom staff members who might lose their jobs because of some of the coal fired power stations reaching the end of their useful lives. It will also create jobs to the locals during different stages of project life cycle.	•
iii. ENVIRONMENTAL IMPACT OF THE NEW GENERATION CAPACITY OF 3 000 MW FROM GAS AT RICHARDS BAY			
1.	Do you think that it is proper for the Energy Regulator to deviate from its mandate and consider other Government Departments mandates when concluding section 34 determinations?	This determination derives its mandate from the Electricity Regulations Act. NERSA does not have powers over mandates of other Government Departments.	
2.	Is it proper for the Energy Regulator to consider matters that will be dealt with at licensing stage during the concurrence process?	This determination derives its mandate from the Electricity Regulations Act. NERSA does not have powers over mandates of other Government Departments.	
IV. PROCUREMENT PROCESS UNDER THE IPP PROCUREMENT PROGRAMME			
1.	Provide your thoughts on Eskom as the chosen procurer of this new generation capacity.		
2.	Provide your thoughts on the method chosen for the procurement of the new generation capacity.		

3.	Section 5 of the Section 34 determination states that the procurement of primary energy and its associated logistic infrastructure must be done in a manner supportive of the State's participation through other State Owned Companies that have the relevant mandate(s). Provide your thoughts on this requirement.	No response Provided	This falls outside of the Regulator's mandate. However, Public (SOEs)-Private-Partnerships (PPPs) are known for producing most desirable outcomes in building mega projects.
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4. Energy Intensive Users Group (EIUG)

No.	Questions from the consultation paper	Stakeholder Comment	NERSA ANALYSIS ON STAKEHOLDER COMMENT
i. TECHNOLOGY			
1.	In your view, is gas the type of technology that is in line with best practice and will the determined capacity ensure security of supply?	Gas is considered a transitional power source towards net-zero goal. For EIUG members who are base-load electricity consumers, gas is a reliable source for its dispatchability especially where intermittent renewable energy is not available. The 3000 MW capacity will assist in ensuring security of supply but on its own is not considered sufficient to ensure security of supply.	- The 3 000MW to be generated from Gas technology is in line with best practice because it will improve the supply-demand imbalance in the country and support the Electricity Regulations Act's objective to promote the use of diverse energy sources and energy efficiency. - It will also support the Government's initiatives to bring affordable, reliable

			and cleaner energy to power consumers.
2.	Provide your comments on the availability of natural gas in South Africa	The gas availability in South Africa is a concern as we are reliant on imported supplies.	Yes, the gas availability in South Africa is a concern. However, Eskom is in a process of engaging with partners like Transnet whose role will be to participate in Liquefied Natural Gas (LNG) receiving infrastructure, regasification, gas transmission and distribution.
3.	Is the determined capacity sufficient to ensure uninterrupted supply of electricity?	The 3000MW capacity will assist in ensuring security of supply but on its own is not considered sufficient.	The electricity system has been constraint for most of the past 14 years. There is also a fundamental change in the energy mix towards renewable resources. Therefore, balanced mid-merit Combined Cycle Power Plant is a technically feasibly solution to help with power crisis.
4.	Provide what you consider to be the risks and challenges associated with this new capacity.	The key risks are the availability of gas and the volatile nature of gas prices over and above the time and cost risks associated with execution.	The risk of the availability of gas can be minimized if Eskom can find local partners to supply reliable gas, at a required pressure and a relatively stable price to avoid volatility of gas

			price. Hence, domestic partners can curb international gas volatility. • Also, gas market aggregation can mitigate this risk whereby a lot of players can compete or use bargaining power to buy it cheaper.
5.	Provide your comments on whether the country will benefit more optimally from gas and the benefits of establishing a gas market.	• There are opportunities for gas beyond electricity production like industrial process heating which could be explored with a well-developed gas market that ensures accessibility and affordability.	• A gas market with domestic partners will curb gas-price volatility caused by the international markets. Therefore, an established gas market in South Africa will ensure accessibility and affordability.
6.	What do you think is the optimal timelines within which this capacity should be made available on the grid, considering the urgency that is required to deal with load shedding in the country?	• A greatly accelerated timeline for additional capacity on the grid is in the National interest. In this context 12-18 months near term horizon should be pursued.	• The optimal timelines are as emphasized in the determination which must not be beyond 2028. • In its feasibility study, Eskom indicated indicative timelines of commissioning different units of which all of the 4 units will be commissioned by no later than the end of year 2028.
ii. COST RECOVERY MECHANISM AND THE IMPACT ON TARIFFS			
1.	Should NERSA continue with the current cost recovery mechanism stipulated in section 13 of the current Multi-Year Price Determination	• The use of a Regulatory Clearing Account (RCA) or a similar mechanism seems to be the most practical recovery mechanism.	• The CRA or any other suitable regulatory tools/mechanism that will be available at that time, will be used for

	Methodology (MYPD) and section 14 of NERSA's CRM Rules? If not, please propose the most suitable cost recovery mechanism for the gas project that would ensure that investment in a gas-to-power facility is economic, resulting in a good opportunity cost for the investor over the life of the asset, while also ensuring that electricity remains affordable.		the 3 000MW of Eskom's gas technology.
2.	Provide your thoughts on the costs that will be associated with establishing the new allocated generation capacity in line with the mandate to ensure the longterm sustainability of the electricity supply industry, as well as affordability and equity.	• No comments were provided by the Stakeholder.	
3.	Provide your thoughts on the impact of adding the additional capacity of 3000MW from gas in relation to NERSA's mandate to ensure the long term sustainability of the electricity supply industry as well as affordability and equity?	• The use of gas will improve energy security and reliability by its virtue of being dispatchable but will negatively affect affordability and price stability due to gas being substantially expensive than coal and its prices highly volatile.	• The sensitivity analysis using correct assumptions will determine their impact on the tariffs. The 3 000MW will provide a mid-merit solution which will provide a technically feasible solution to help with power crisis.

4.	What will be the anticipated tariff impact to customers or end users in R/KWh if 3000MW of gas to power is produced?	Without the necessary research, modelling tools and time, the EIUG is unable to quantify this this tariff impact. However, it is acknowledged that the impact will be to increase the prices compared to coal, nuclear and renewable energy.	The revenue requirement or any other price determination methodology that will be available at that time, will be used to calculate the tariff impact of 3 000MW.
5.	Provide your comments on the socio-economic impact of the determined capacity in terms of the number of jobs that can be created, including the objective of a just transition from coal to clean technologies.	The EIUG does not have numbers but it believes this capacity will create jobs both during construction and operations. It may also significantly contribute to catalysing a fully-fledged gas market in the country which will bring more jobs and economic benefits. So there will be an economic benefit to the country. This capacity will have a positive impact in reducing grid carbon intensity.	- Since the Electricity Supply Industry (ESI) is transitioning in line with Just Energy Transition, the 3 000MW from gas technology will minimize the impact of Eskom staff members who might lose their jobs because of some of the coal fired power stations reaching the end of their useful lives. - It will also create jobs to the locals during different stages of project life cycle.
iii. ENVIRONMENTAL IMPACT OF THE NEW GENERATION CAPACITY OF 3 000 MW FROM GAS AT RICHARDS BAY			
1.	Do you think that it is proper for the Energy Regulator to deviate from its mandate and consider other Government Departments mandates when concluding section 34 determinations?	No. However where there is an official policy specific matter NERSA may have to consider such. As an example, NERSA will have to check the legal validity of this determination considering that a gas determination for IRP2019 gas allocation was already done in September 2020	This determination must derive its mandate from the Electricity Regulations Act. NERSA does not have powers over mandates of other Government Departments.
2.	Is it proper for the Energy Regulator to consider	Not proper.	Yes it is not proper.

	matters that will be dealt with at licensing stage during the concurrence process?		
IV. PROCUREMENT PROCESS UNDER THE IPP PROCUREMENT PROGRAMME			
1.	Provide your thoughts on Eskom as the chosen procurer of this new generation capacity.	Supported however due consideration should have been given to private sector procurement given the precarious nature of Eskom balance sheet and large scale projects execution challenges.	
2.	Provide your thoughts on the method chosen for the procurement of the new generation capacity.	Supported	
3.	Section 5 of the Section 34 determination states that the procurement of primary energy and its associated logistic infrastructure must be done in a manner supportive of the State's participation through other State Owned Companies that have the relevant mandate(s). Provide your thoughts on this requirement.	Supported	

5. GIGAJoule INTERNATIONAL

No.	Questions from the consultation paper	Stakeholder Comment	NERSA ANALYSIS ON STAKEHOLDER COMMENT
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i. TECHNOLOGY			
1.	In your view, is gas the type of technology that is in line with best practice and will the determined capacity ensure security of supply?	Natural gas technology is also seen as the best solution to decrease coal dependence in the short term and reach net-zero milestones – not only does natural gas alone reduce carbon emissions compared to coal-generated plants by circa 60%, but it enables renewable energy projects the effect is a total reduction of closer to 80%.	- The 3 000MW to be generated from Gas technology is in line with best practice because it will improve the supply-demand imbalance in the country and support the Electricity Regulations Act's objective to promote the use of diverse energy sources and energy efficiency. - It will also support the Government's initiatives to bring affordable, reliable and cleaner energy to power consumers.
2.	Provide your comments on the availability of natural gas in South Africa	The gas is available in South Africa, ROMCO started transporting 120million GJ/a in 2004 for Sasol's consumption and inland customers. Since 2004, the quantity of natural gas supplied via ROMPCO increased to 180million GJ/a where it reached a cap due to the limitations of the Pande and Temane gas fields. Due to this current cap, there is significant pent-up demand for natural gas.	Yes, the gas availability in South Africa is a concern. However, Eskom is in a process of engaging with partners like Transnet whose role will be to participate in Liquefied Natural Gas (LNG) receiving infrastructure, regasification, gas transmission and distribution.
3.	Is the determined capacity sufficient to ensure uninterrupted supply of electricity?	The current supply sources are insufficient, and the risks related to relying solely on the Richards Bay project require mitigation.	The electricity system has been constraint for most of the past 14 years. There is also a fundamental change in the energy mix towards renewable resources. Therefore,

			balanced mid-merit Combined Cycle Power Plant is a technically feasible solution to help with power crisis.
4.	Provide what you consider to be the risks and challenges associated with this new capacity.	- Risks and challenges associated with developing new gas-to-power capacity in South Africa include some of the following: i) Financing of the infrastructure will require robust supporting off-take and fuel supply agreements with relevant Take-or-Pay commitments. ii) Aggregating sufficient offtake to bank new LNG importation and natural gas supply. iii) Securing long-term LNG supply contracts with reliable and reputable LNG supplier(s). iv) The current impact of the Russia-Ukraine war on the international gas supply market.	- The risk of the availability of gas can be minimized if Eskom can find local partners to supply reliable gas, at a required pressure and a relatively stable price to avoid volatility of gas price. Hence, domestic partners can curb international gas volatility. - Also, gas market aggregation can mitigate this risk whereby a lot of players can compete or use bargaining power to buy it cheaper.
5.	Provide your comments on whether the country will benefit more optimally from gas and the benefits of establishing a gas market.	The gas market will stimulate the availability of cheaper and cleaner natural gas. This will result into growth of the industrial market as investment increased. The supply of natural gas to the industrial market grew up to 40PJ/a for the past 15 years.	A gas market with domestic partners will curb gas-price volatility caused by the international markets. Therefore, an established gas market in South Africa will ensure accessibility and affordability.
6.	What do you think is the optimal timelines within which this capacity should be made available on the grid, considering the urgency that is required to deal with load shedding in the country?	The addition of more power generation is a matter of urgency. Combined Cycle Gas Turbines (CCGTs) are known to have a much shorter development and construction period and can be completed within 24 to 30 months after financial approval is reached and at a much lower capital cost than coal and nuclear power stations.	- The optimal timelines are as emphasized in the determination which must not be beyond 2028. - In its feasibility study, Eskom indicated indicative timelines of commissioning different units of which all of the 4 units

			will be commissioned by no later than the end of year 2028.
ii. COST RECOVERY MECHANISM AND THE IMPACT ON TARIFFS			
1.	Should NERSA continue with the current cost recovery mechanism stipulated in section 13 of the current Multi-Year Price Determination Methodology (MYPD) and section 14 of NERSA's CRM Rules? If not, please propose the most suitable cost recovery mechanism for the gas project that would ensure that investment in a gas-to-power facility is economic, resulting in a good opportunity cost for the investor over the life of the asset, while also ensuring that electricity remains affordable.	Gigajoule believes that the pricing structure should be simplified and provide for investors to interact with NERSA to finalize the price to ensure fair and competitive cost of capital recovery and electricity.	The CRA or any other suitable regulatory tools/mechanism that will be available at that time, will be used for the 3 000MW of Eskom's gas technology.
2.	Provide your thoughts on the costs that will be associated with establishing the new allocated generation capacity in line with the mandate to ensure the longterm sustainability of the electricity supply	The overnight cost for CCGT in US Dollar is 0.85b/GW.	EPRI and Lazard provide the overnight cost of 3 000MW produced from gas. Although these costs are indicative but they provide an indication how much it will cost to produce 3 000MW from gas.

	industry, as well as affordability and equity.		
3.	How should the cost associated with the fuel for this technology be dealt with within the cost recovery mechanism in order to mitigate against rapid changes in prices of gas?	- To sign long-term agreements and pricing which: i) Allows for a larger fixed component. ii) Decrease exposure to the volatility of the short-term spot market. iii) Pricing formulae that consider alternative fuel supply indices.	The CRA or any other regulatory price methodology that will be available at that time will be used to mitigate against rapid changes in prices of gas.
4.	What will be the anticipated tariff impact to customers or end users in R/kWh if 3000MW of gas to power is produced?	The net benefit to the customer/end-user can be realized when using a blended LCOE rather than having 3 000MW's LCOE calculated on a stand-alone basis.	The revenue requirement or any other price determination methodology that will be available at that time, will be used to calculate the tariff impact of 3 000MW.
5.	Provide your comments on the socio-economic impact of the determined capacity in terms of the number of jobs that can be created, including the objective of a just transition from coal to clean technologies.	- Natural gas is the enabler of two main factors: i) Just transition away from coal to clean renewables. ii) Gas is available to industry for economic growth and job creation. iii) Gas-fired power stations and the gas infrastructure value chain in itself do not create the number of jobs required to alleviate the employment crisis. Mostly skilled workers are employed during the operation phase, with less than 100 people for a 2 000MW power plant.	- Since the Electricity Supply Industry (ESI) is transitioning in line with Just Energy Transition, the 3 000MW from gas technology will minimize the impact of Eskom staff members who might lose their jobs because of some of the coal fired power stations reaching the end of their useful lives. - It will also create jobs to the locals during different stages of project life cycle.
iii. ENVIRONMENTAL IMPACT OF THE NEW GENERATION CAPACITY OF 3 000 MW FROM GAS AT RICHARDS BAY			

1.	Do you think that it is proper for the Energy Regulator to deviate from its mandate and consider other Government Departments mandates when concluding section 34 determinations?	The Stakeholder did not respond directly to the question asked.	
2.	Is it proper for the Energy Regulator to consider matters that will be dealt with at licensing stage during the concurrence process?	The Stakeholder did not respond directly to the question asked.	
IV. PROCUREMENT PROCESS UNDER THE IPP PROCUREMENT PROGRAMME			
1.	Provide your thoughts on Eskom as the chosen procurer of this new generation capacity.	The Stakeholder did not respond directly to the question asked.	
2.	Provide your thoughts on the method chosen for the procurement of the new generation capacity.	The Stakeholder did not respond directly to the question asked.	
3.	Section 5 of the Section 34 determination states that the procurement of primary energy and its associated logistic infrastructure must be done in a manner supportive of the State's participation through other State Owned Companies that have the relevant mandate(s).	The Stakeholder did not respond directly to the question asked.	

	Provide your thoughts on this requirement.		
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6. Meridian Economics

No.	Questions from the consultation paper	Stakeholder Comment	NERSA ANALYSIS ON STKEHOLDER COMMENT
i. TECHNOLOGY			
19.	In your view, is gas the type of technology that is in line with best practice and will the determined capacity ensure security of supply?	- Independent studies are unanimous on the optimal role for gas-to-power in South Africa – the optimal role is as peaking support. i.e., one tenth of the large-scale role that CCGT is typically installed to fulfil - CCGT gas will displace new RE capacity which, combined with OCGTs or other peaking support technology, would achieve security of supply at much lower cost with much lower emissions than a CCGT option. - For at least the next fifteen years, until the bulk of the coal is off the system, large gas-to-power plants are not chosen by the power system models to generate significant quantities of energy - If significant CCGT capacity is built in the next decade in conjunction with the large renewable capacity required to fulfil least cost build requirements, and the CCGT capacity is dispatched in an economically rational manner in accordance with a least cost merit order, the CCGT capacity will stand idle much of the time. It would	- The 3 000MW to be generated from Gas technology is in line with best practice because it will improve the supply-demand imbalance in the country and support the Electricity Regulations Act's objective to promote the use of diverse energy sources and energy efficiency. - It will also support the Government's initiatives to bring affordable, reliable and cleaner energy to power consumers.

		therefore be a waste of money to install CCGT plant in the timeframe envisaged in the concurrence.	
20.	Provide your comments on the availability of natural gas in South Africa	• There is virtually no domestic natural gas available in South Africa • Gas-fired power would need to source its fuel from LNG for the time frame being considered in this decision • LNG is not necessarily cheaper than diesel for firing peaking power plant (that is required by the power system) and in fact likely to be more expensive than diesel for the small quantities needed for peaking.	• The gas availability in South Africa is a concern. However, Eskom is in a process of engaging with partners like Transnet whose role will be to participate in Liquefied Natural Gas (LNG) receiving infrastructure, regasification, gas transmission and distribution.
21.	Is the determined capacity sufficient to ensure uninterrupted supply of electricity?	• This question cannot be answered in isolation and depends on the other resources that are planned to be brought on line during the same period. All independent power modelling studies show that security of supply is best achieved through a combination of substantial RE build and peaking plant capacity.	• The electricity system has been constraint for most of the past 14 years. There is also a fundamental change in the energy mix towards renewable resources. Therefore, balanced mid-merit Combined Cycle Power Plant is a technically feasibly solution to help with power crisis.
22.	Provide what you consider to be the risks and challenges associated with this new capacity.	• The viable use of gas in the power sector requires larger anchor gas demand from other sectors at each supply point in order to be economically feasible. • Forcing large-scale gas use into the power generation portfolio in South Africa instead of the much smaller	• The risk of the availability of gas can be minimized if Eskom can find local partners to supply reliable gas, at a required pressure and a relatively stable price to avoid volatility of gas

		alternative peaking role to support renewables will increase the cost of the electricity so generated by more than 40%. It will increase emissions seven-fold compared to the alternative whilst reducing the socio-economic benefit that would accrue to the country. (see section 4.3 of our “Hot air about Gas” report [5])	price. Hence, domestic partners can curb international gas volatility. • Also, gas market aggregation can mitigate this risk whereby a lot of players can compete or use bargaining power to buy it cheaper.
23.	Provide your comments on whether the country will benefit more optimally from gas and the benefits of establishing a gas market.	• The use of gas in the power sector must be motivated by the economics of using gas to generate electricity in South Africa in the context of the other generation alternatives. Using the power sector to cross-subsidise otherwise sub-economic use of gas elsewhere in the economy is bad economics and will overall reduce the wealth of the country.	• A gas market with domestic partners will curb gas-price volatility caused by the international markets. Therefore, an established gas market in South Africa will ensure accessibility and affordability.
24.	What do you think is the optimal timelines within which this capacity should be made available on the grid, considering the urgency that is required to deal with load shedding in the country?	• The grid primarily requires energy as a matter of urgency. The quickest route to increasing energy available on the grid is to accelerate the rollout of renewable technologies. The grid also requires peaking plant as a matter of urgency to support short-term interruptions due to failure of the coal plant. However, there is no need for CCGT plant on the grid in the next decade at least.	• The optimal timelines are as emphasized in the determination which must not be beyond 2028. • In its feasibility study, Eskom indicated indicative timelines of commissioning different units of which all of the 4 units will be commissioned by no later than the end of year 2028.
ii. COST RECOVERY MECHANISM AND THE IMPACT ON TARIFFS			
1.	Should NERSA continue with the current cost recovery mechanism stipulated in section 13 of the current Multi-Year Price Determination	• No, NERSA should develop the most suitable cost recovery mechanism for the gas project that would ensure that investment in a gas-to-power facility is economic, resulting in a good opportunity cost for the investor over	• Cost Recovery Mechanism should have not been included as part of section 34 determination as it falls

	Methodology (MYPD) and section 14 of NERSA's CRM Rules? If not, please propose the most suitable cost recovery mechanism for the gas project that would ensure that investment in a gas-to-power facility is economic, resulting in a good opportunity cost for the investor over the life of the asset, while also ensuring that electricity remains affordable.	the life of the asset, while also ensuring that electricity remains affordable.	within the scope of the MYPD4 Methodology and CRM Rules.
2.	Provide your thoughts on the costs that will be associated with establishing the new allocated generation capacity in line with the mandate to ensure the longterm sustainability of the electricity supply industry, as well as affordability and equity.	• Refer to section 4 of our gas report "Hot Air about Gas" [5]. A decision to concur with installation of CCGT power will undermine the long-term sustainability of the ESI and expose consumers to excessive prices, and ongoing price volatility on a substantial portion of the generation costs. It will increase South Africa's reliance on imported fuels.	• EPRI and Lazard provide the overnight cost of 3 000MW produced from gas. Although these costs are indicative but they provide an indication how much it will cost to produce 3 000MW from gas.
3.	How should the cost associated with the fuel for this technology be dealt with within the cost recovery mechanism in order to mitigate against rapid changes in prices of gas?	• The best mitigation against the price volatility wrought on the electricity system by gas prices, is to minimize the amount of electricity that is generated from gas. This is borne out by the results of power system modelling which deploy gas only in a peaking/support role at very low capacity factors.	• The CRA or any other regulatory price methodology that will be available at that time will be used to mitigate against rapid changes in prices of gas.

4.	What will be the anticipated tariff impact to customers or end users in R/kWh if 3 000MW of gas-to-power is procured?	At current prices of gas and renewable energy, the cost of power from CCGT is almost double the cost of equivalent power from a combination of renewables and peaking plant. Thus a decision to proceed with the CCGT procurement will likely result in tariffs for that portion of power that are nearly double the level that they could be if the more rational combination of renewables and peaking plant were installed.	The revenue requirement or any other price determination methodology that will be available at that time, will be used to calculate the tariff impact of 3 000MW.
5.	Provide your comments on the socio-economic impact of the determined capacity in terms of the number of jobs that can be created, including the objective of a just transition from coal to clean technologies.	There is substantial opportunity to localize large portions of the RE value chain, whereas there is very little opportunity to localize any of the value chain of largescale gas power generation. Thus, a decision to procure 3GW of CCGT will not only result in higher electricity prices with associated dampening on economic growth and employment, but directly result in the elimination of jobs that would have been created had the same power been provided (at lower cost) by a combination of RE and peaking plant.	- Since the Electricity Supply Industry (ESI) is transitioning in line with Just Energy Transition, the 3 000MW from gas technology will minimize the impact of Eskom staff members who might lose their jobs because of some of the coal fired power stations reaching the end of their useful lives. - It will also create jobs to the locals during different stages of project life cycle.
iii. ENVIRONMENTAL IMPACT OF THE NEW GENERATION CAPACITY OF 3 000 MW FROM GAS AT RICHARDS BAY			
1.	Do you think that it is proper for the Energy Regulator to deviate from its mandate and consider other Government Departments mandates when concluding section 34 determinations?	(Please refer to section 3.3, 4.2 and 4.3 of our “Hot air about Gas” report [5])	This determination derives its mandate from the Electricity Regulations Act. NERSA does not have powers over mandates of other Government Departments.

2.	Is it proper for the Energy Regulator to consider matters that will be dealt with at licensing stage during the concurrence process?	(Please refer to section 3.3, 4.2 and 4.3 of our “Hot air about Gas” report [5])	This determination derives its mandate from the Electricity Regulations Act. NERSA does not have powers over mandates of other Government Departments.
IV. PROCUREMENT PROCESS UNDER THE IPP PROCUREMENT PROGRAMME			
1.	Provide your thoughts on Eskom as the chosen procurer of this new generation capacity.	No Comment Provided	
2.	Provide your thoughts on the method chosen for the procurement of the new generation capacity.	No Comment Provided	
3.	Section 5 of the Section 34 determination states that the procurement of primary energy and its associated logistic infrastructure must be done in a manner supportive of the State’s participation through other State Owned Companies that have the relevant mandate(s). Provide your thoughts on this requirement.	- There is no reason that is given as to why this requirement is necessary. - Why should one particular entity be advantaged over others? - This will amount to uncompetitive procurement in terms of the constitution and the PFMA. - This appears to be unlawful as no rational and reasonable basis has been provided for NERSA to concur with this requirement. Without this, NERSA cannot lawfully concur.	This falls outside of the Regulator’s mandate. However, Public (SOEs)-Private-Partnerships (PPPs) are known for producing most desirable outcomes in building mega projects.
V. CONCURRENCE PROCESS TO THE MINISTERIAL DETERMINATIONS WITH INDICATIVE TIMELINES			
		The EIA for the 3GW CCGT plant is currently being challenged in court. This will delay the process in any event providing ample time for NERSA to hold a public hearing.	

		There is no reason for the extreme haste of this public process, and it fundamentally prejudices stakeholder rights and NERSA's ability to take a lawful decision in the public interest.	
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7. MINERALS COUNCIL OF SOUTH AFRICA

No.	Questions from the consultation paper	Stakeholder Comment	NERSA ANALYSIS ON STAKEHOLDER COMMENT
i. TECHNOLOGY			
1.	In your view, is gas the type of technology that is in line with best practice and will the determined capacity ensure security of supply?	The Stakeholder did not respond directly to the question asked.	
2.	Provide your comments on the availability of natural gas in South Africa	The difficulty in South Africa is that there has been little exploration undertaken not just in mineral exploration but also gas. Gas resources and reserves are certainly available. The DMRE must develop an investor friendly exploration plan. This means the following: a) Certainty – It has been proven that the constant tweaking of the Mining Charter introduces uncertainty in the operating environment. b) MPRDA/EIA – Botswana's mineral and energy policies have remained unchanged with the former having been promulgated in 1999 and has barely	Yes, the gas availability in South Africa is a concern. However, Eskom is in a process of engaging with partners like Transnet whose role will be to participate in Liquefied Natural Gas (LNG) receiving infrastructure, regasification, gas transmission and distribution.

		been amended. It is not surprising that Botswana, a way smaller country than South Africa, is the best mining jurisdiction in Africa in terms of investor stability. c) Civil society have a lot of power through the constitution and the court, and they have been behind the blocking of many an exploration endeavor.	
3.	Is the determined capacity sufficient to ensure uninterrupted supply of electricity?	The 3000MW capacity is adequate given the gap between electricity demand and supply in SA.	The electricity system has been constraint for most of the past 14 years. There is also a fundamental change in the energy mix towards renewable resources. Therefore, balanced mid-merit Combined Cycle Power Plant is a technically feasibly solution to help with power crisis.
4.	Provide what you consider to be the risks and challenges associated with this new capacity.	Supply risks – Where will SA get the gas from? The Gas Masterplan Base case report published by the DMRE in 2021 states the following: “Three supply sources of natural gas are available to South Africa: a) Piped natural gas from neighbouring countries. b) Imported LNG; and c) Domestic natural gas, either from conventional or unconventional reserves.”	- The risk of the availability of gas can be minimized if Eskom can find local partners to supply reliable gas, at a required pressure and a relatively stable price to avoid volatility of gas price. Hence, domestic partners can curb international gas volatility. - Also, gas market aggregation can mitigate this risk whereby a lot of players can compete or use bargaining power to buy it cheaper.

		The first two supply sources are uncertain. The last one is not going to happen unless business friendly policies are promulgated, including curtailing the influence of civil society in the implementation of economic policies and projects.	
5.	Provide your comments on whether the country will benefit more optimally from gas and the benefits of establishing a gas market.	The Gas Masterplan Base case report indicates that gas will only be 2.6% of the energy mix. For such an important energy carrier, that is too little to change South Africa's growth and job creation prospects.	A gas market with domestic partners can curb gas-price volatility caused by the international markets. Therefore, an established gas market in South Africa will ensure accessibility and affordability.
6.	What do you think is the optimal timelines within which this capacity should be made available on the grid, considering the urgency that is required to deal with load shedding in the country?	While OCGTs and CCGT (open/combined cycle gas turbines) are cheap to build and good for the short-run in producing energy they still take three to four years to commission. South Africa's electricity requirement has to be addressed quite urgently.	- The optimal timelines are as emphasized in the determination which must not be beyond 2028. - In its feasibility study, Eskom indicated indicative timelines of commissioning different units of which all of the 4 units will be commissioned by no later than the end of year 2028.
ii. COST RECOVERY MECHANISM AND THE IMPACT ON TARIFFS			
1.	Should NERSA continue with the current cost recovery mechanism stipulated in section 13 of the current Multi-Year Price Determination Methodology (MYPD) and	NERSA should continue with the current cost recovery mechanism stipulated in section 13 of the current Multi-Year Price Determination Methodology (MYPD) and section 14 of NERSA's CRM Rules	The CRA or any other suitable regulatory tools/mechanism that will be available at that time, will be used for the 3 000MW of Eskom's gas technology.

	section 14 of NERSA's CRM Rules? If not, please propose the most suitable cost recovery mechanism for the gas project that would ensure that investment in a gas-to-power facility is economic, resulting in a good opportunity cost for the investor over the life of the asset, while also ensuring that electricity remains affordable.		
2.	Provide your thoughts on the costs that will be associated with establishing the new allocated generation capacity in line with the mandate to ensure the longterm sustainability of the electricity supply industry, as well as affordability and equity.	• Cost of the feedstock, which will be impacted by the exchange rate • Cost of generation, transmission and distribution • Maintenance cost of generation, transmission, and distribution • Other costs incl. labour • Markup (return on investment, ROI)	• EPRI and Lazard provide the overnight cost of 3 000MW produced from gas. Although these costs are indicative but they provide an indication how much it will cost to produce 3 000MW from gas.
3.	Provide your thoughts on the impact of adding the additional capacity of 3000MW from gas in relation to NERSA's mandate to ensure the long term sustainability of the electricity	• Affordability is the main issue since the gas must be procured in US dollars. With the Rand so volatile, price certainty will be a difficult thing. We have to look at the price of liquid fuels to understand the extent of the	• The sensitivity analysis using correct assumptions will determine their impact on the tariffs. The 3 000MW will provide a mid-merit solution which

	supply industry as well as affordability and equity?	volatility that the electricity market will be subjected to because of having to import the gas.	will provide a technically feasible solution to help with power crisis.
4.	How should the cost associated with the fuel for this technology be dealt with within the cost recovery mechanism in order to mitigate against rapid changes in prices of gas?	- This is a complex exercise as it involves hedging on two fronts: a) hedging the against the depreciation of the rand against the b) US dollar; and hedging against the international gas price volatility. - Minerals Council's view is that without significant domestic production of gas, price volatility of gas generated electricity will be inevitable. A clear example is the developments in the petroleum and gas markets which have been affected by the invasion of Ukraine by Russia.	The CRA or any other regulatory price methodology that will be available at that time will be used to mitigate against rapid changes in prices of gas.
5.	What will be the anticipated tariff impact to customers or end users in R/kWh if 3000MW of gas to power is produced?	The Stakeholder did not respond to the question asked.	
6.	Provide your comments on the socio-economic impact of the determined capacity in terms of the number of jobs that can be created, including the	Natural gas is the cleanest fossil fuel to use. It is also reliable and has lesser CO2 emissions. The energy sector, including OCGTs and CCGTs, is capital intensive. Add to that the fact that the gas will predominantly be imported, it simply means that most	Since the Electricity Supply Industry (ESI) is transitioning in line with Just Energy Transition, the 3 000MW from gas technology will minimize the impact of Eskom staff members who

	objective of a just transition from coal to clean technologies.	of the jobs, or at least an equal share, will be created in countries where the gas is sourced.	might lose their jobs because of some of the coal fired power stations reaching the end of their useful lives. • It will also create jobs to the locals during different stages of project life cycle.
iii. ENVIRONMENTAL IMPACT OF THE NEW GENERATION CAPACITY OF 3 000 MW FROM GAS AT RICHARDS BAY			
1.	Do you think that it is proper for the Energy Regulator to deviate from its mandate and consider other Government Departments mandates when concluding section 34 determinations?	• No. An ideal model, though, would be to have NERSA as a one-stop shop for energy (electricity and liquid fuels) applications. It is unlikely that this would conflate the role of the regulator and that of the DMRE. Our experience is that other government departments tend to have a strict application of the 'rules' thus stifling investment in the sector.	• This determination derives its mandate from the Electricity Regulations Act. NERSA does not have powers over mandates of other Government Departments.
2.	Is it proper for the Energy Regulator to consider matters that will be dealt with at licensing stage during the concurrence process?	(Please refer to section 3.3, 4.2 and 4.3 of our "Hot air about Gas" report [5])	• This determination derives its mandate from the Electricity Regulations Act. NERSA does not have powers over mandates of other Government Departments.
2.	Is it proper for the Energy Regulator to consider matters that will be dealt with at licensing stage during the concurrence process?	• Ideally that should be the case. NERSA should be a one-stop shop as stated above.	• Minerals Council suggest a change in policy which falls outside of the ambit of NERSA.

IV. PROCUREMENT PROCESS UNDER THE IPP PROCUREMENT PROGRAMME			
1.	Provide your thoughts on Eskom as the chosen procurer of this new generation capacity.	• Eskom should not be part of this process as it presents a conflict of interest for them. Currently, in the absence of the ISMO, Eskom can be the chosen procurer.	• This determination is premised on the Eskom's application. Hence, it will be processed as such
2.	Provide your thoughts on the method chosen for the procurement of the new generation capacity.	• The REIPPPP is rated as one of the best in the world because of its transparency among other factors. This has to continue going forward.	• This determination is premised on the Eskom's application. Hence, it will be processed as such
3.	Section 5 of the Section 34 determination states that the procurement of primary energy and its associated logistic infrastructure must be done in a manner supportive of the State's participation through other State Owned Companies that have the relevant mandate(s). Provide your thoughts on this requirement.	South Africa is in the situation of inadequate energy supply and lack of infrastructure maintenance because of government's insistence, through SOEs, of wanting to be a player in the ESI. In time government's competing interests and objectives ensure that its financial resources are stretched, rendering it incapable of attending to areas where it is invested. In short, the industry should be opened up to the private sector. The liberalisation should be right through the ESI value chain. Where natural monopolies are inevitable the concessionaire model should be applied e.g., transmission and distribution. This will free up state resources to focus on social investment initiatives such as health, education, and policing/security. State resources would also go into addressing market failures	• This falls outside of the Regulator's mandate. However, Public (SOEs)-Private-Partnerships (PPPs) are known for producing most desirable outcomes in building mega projects.

8. Moondream

No.	Questions from the consultation paper	Stakeholder Comment	NERSA ANALYSIS ON STKEHOLDER COMMENT
i. TECHNOLOGY			
25.	In your view, is gas the type of technology that is in line with best practice and will the determined capacity ensure security of supply?	The stakeholder did not respond to the question asked.	- The 3 000MW to be generated from Gas technology is in line with best practice because it will improve the supply-demand imbalance in the country and support the Electricity Regulations Act's objective to promote the use of diverse energy sources and energy efficiency. - It will also support the Government's initiatives to bring affordable, reliable and cleaner energy to power consumers. - However, this analysis provided above does not take away the fact that this is not a section 34 determination. Therefore, another process to facilitate this application should be followed
26.	Provide your comments on the availability of natural gas in South Africa	Regional natural gas supply from neighbouring countries is constrained and supply declining. When these conditions change, South Africa should look to secure supply from neighbouring countries.	The gas availability in South Africa is a concern. However, Eskom is in a process of engaging with partners like Transnet whose role will be to participate in Liquefied Natural Gas

			(LNG) receiving infrastructure, regasification, gas transmission and distribution.
27.	Is the determined capacity sufficient to ensure uninterrupted supply of electricity?	• The proposed 3 000MW gas to power plant running at full load for 24 hours will result in 1 000MW as the country needs at least 4 000MW to combat load shedding. • Therefore, an additional 1 000MW to 1 500MW allocated to gas will be sufficient to ensure uninterrupted supply of electricity.	• The electricity system has been constraint for most of the past 14 years. There is also a fundamental change in the energy mix towards renewable resources. Therefore, balanced mid-merit Combined Cycle Power Plant is a technically feasibly solution to help with power crisis.
28.	Provide what you consider to be the risks and challenges associated with this new capacity.	• The availability of natural gas is a challenge in the short term, but global analysts' consensus is that the supply side constraint will stabilize over the next few years. • Ability to move natural gas around the country will prove challenging due to limited and aged gas pipeline infrastructure in South Africa.	• The risk of the availability of gas can be minimized if Eskom can find local partners to supply reliable gas, at a required pressure and a relatively stable price to avoid volatility of gas price. Hence, domestic partners can curb international gas volatility. • Also, gas market aggregation can mitigate this risk whereby a lot of players can compete or use bargaining power to buy it cheaper.
29.	Provide your comments on whether the country will benefit more optimally from gas and the benefits of establishing a gas market.	• The key to this question is the load factor of the gas power plant, the resulting volumes will provide the catalyst for gas investment infrastructure and also allow for the expansion of the gas market to other users.	• A gas market with domestic partners will curb gas-price volatility caused by the international markets. Therefore, an established gas market in South

			Africa will ensure accessibility and affordability.
30.	What do you think is the optimal timelines within which this capacity should be made available on the grid, considering the urgency that is required to deal with load shedding in the country?	No comment was provided by the stakeholder on this question.	- The optimal timelines are as emphasized in the determination which must not be beyond 2028. - In its feasibility study, Eskom indicated indicative timelines of commissioning different units of which all of the 4 units will be commissioned by no later than the end of year 2028.
ii. COST RECOVERY MECHANISM AND THE IMPACT ON TARIFFS			
1.	Should NERSA continue with the current cost recovery mechanism stipulated in section 13 of the current Multi-Year Price Determination Methodology (MYPD) and section 14 of NERSA's CRM Rules? If not, please propose the most suitable cost recovery mechanism for the gas project that would ensure that investment in a gas-to-power facility is economic, resulting in a good opportunity cost for the investor over the life of the asset, while also	It will be reasonable to have a retail tariff component of the annual price adjustment to cater for the movements in the gas pricing and fluctuations in currency related to the gas purchases for gas to power and peaking plants.	Cost Recovery Mechanism should have not been included as part of section 34 determination as it falls within the scope of the MYPD4 Methodology and CRM Rules.

	ensuring that electricity remains affordable.		
2.	Provide your thoughts on the costs that will be associated with establishing the new allocated generation capacity in line with the mandate to ensure the longterm sustainability of the electricity supply industry, as well as affordability and equity.	The capital cost for a gas to power plant is currently in the region of \$850/kW for CCGT plant.	EPRI and Lazard provide the overnight cost of 3 000MW produced from gas. Although these costs are indicative but they provide an indication how much it will cost to produce 3 000MW from gas.
3.	How should the cost associated with the fuel for this technology be dealt with within the cost recovery mechanism in order to mitigate against rapid changes in prices of gas?	It will be reasonable to have a retail tariff component of the annual price adjustment to cater for the movements in the gas pricing and fluctuations in currency related to the gas purchases for gas to power and peaking plants.	The CRA or any other regulatory price methodology that will be available at that time will be used to mitigate against rapid changes in prices of gas.
4.	What will be the anticipated tariff impact to customers or end users in R/kWh if 3 000MW of gas-to-power is procured?	The tariff will range from R3/kWh to R4/kWh	The revenue requirement or any other price determination methodology that will be available at that time, will be used to calculate the tariff impact of 3 000MW.
5.	Provide your comments on the socio-economic impact of the determined capacity in terms of the number of jobs that can be created, including the objective of a just	This project will act as a catalyst in job creation, energy security, supplier development and local equity.	Since the Electricity Supply Industry (ESI) is transitioning in line with Just Energy Transition, the 3 000MW from gas technology will minimize the impact of Eskom staff members who might lose their jobs because of some

	transition from coal to clean technologies.		of the coal fired power stations reaching the end of their useful lives. • It will also create jobs to the locals during different stages of project life cycle.
iii. ENVIRONMENTAL IMPACT OF THE NEW GENERATION CAPACITY OF 3 000 MW FROM GAS AT RICHARDS BAY			
1.	Do you think that it is proper for the Energy Regulator to deviate from its mandate and consider other Government Departments mandates when concluding section 34 determinations?	• At a concurrence stage, NERSA should not deviate from its mandate	• This determination derives its mandate from the Electricity Regulations Act. NERSA does not have powers over mandates of other Government Departments.
2.	Is it proper for the Energy Regulator to consider matters that will be dealt with at licensing stage during the concurrence process?	• At a concurrence stage, NERSA should not deviate from its mandate	• This determination derives its mandate from the Electricity Regulations Act. NERSA does not have powers over mandates of other Government Departments.
IV. PROCUREMENT PROCESS UNDER THE IPP PROCUREMENT PROGRAMME			
1.	Provide your thoughts on Eskom as the chosen procurer of this new generation capacity.	• Eskom should only be allowed to be a Buyer but not a Procurer.	• The chosen Procurer falls outside of the NERSA mandate.
2.	Provide your thoughts on the method chosen for the procurement of the new generation capacity.	• Eskom should only be allowed to be a Buyer but not a Procurer.	• Procurement process falls outside of the NERSA mandate. These powers are for other Government Departments to execute.

3.	Section 5 of the Section 34 determination states that the procurement of primary energy and its associated logistic infrastructure must be done in a manner supportive of the State's participation through other State Owned Companies that have the relevant mandate(s). Provide your thoughts on this requirement.		This falls outside of the Regulator's mandate. However, Public (SOEs)-Private-Partnerships (PPPs) are known for producing most desirable outcomes in building mega projects.
V. CONCURRENCE PROCESS TO THE MINISTERIAL DETERMINATIONS WITH INDICATIVE TIMELINES			

9. Phakwe Group

No.	Questions from the consultation paper	Stakeholder Comment	NERSA ANALYSIS ON STKEHOLDER COMMENT
i. TECHNOLOGY			
31.	In your view, is gas the type of technology that is in line with best practice and will the determined capacity ensure security of supply?	The presence of gas technology is critical in ensuring security of supply in South Africa.	- The 3 000MW to be generated from Gas technology is in line with best practice because it will improve the supply-demand imbalance in the country and support the Electricity Regulations Act's objective to promote the use of diverse energy sources and energy efficiency. - It will also support the Government's initiatives to bring affordable, reliable

			and cleaner energy to power consumers. • However, this analysis provided above does not take away the fact that this is not a section 34 determination. Therefore, another process to facilitate this application should be followed
32.	Provide your comments on the availability of natural gas in South Africa	• There is currently no sufficient natural gas present in South Africa to supply this generation capacity or any other natural gas demand in this country.	• The gas availability in South Africa is a concern. However, Eskom is in a process of engaging with partners like Transnet whose role will be to participate in Liquefied Natural Gas (LNG) receiving infrastructure, regasification, gas transmission and distribution.
33.	Is the determined capacity sufficient to ensure uninterrupted supply of electricity?	• No, the current and the forecasted EAF from Eskom coal fleet indicates the gap in Eskom's own generation from 4GW to 6GW.	• The electricity system has been constraint for most of the past 14 years. There is also a fundamental change in the energy mix towards renewable resources. Therefore, balanced mid-merit Combined Cycle Power Plant is a technically feasibly solution to help with power crisis.
34.	Provide what you consider to be the risks and challenges associated with this new capacity.	• Gas to power plants require a solid natural gas supply infrastructure. Therefore, the most significant challenge	• The risk of the availability of gas can be minimized if Eskom can find local partners to supply reliable gas, at a

		is to ensure that the supply chain offer of natural gas matches the demand requirements.	required pressure and a relatively stable price to avoid volatility of gas price. Hence, domestic partners can curb international gas volatility. Also, gas market aggregation can mitigate this risk whereby a lot of players can compete or use bargaining power to buy it cheaper.
35.	Provide your comments on whether the country will benefit more optimally from gas and the benefits of establishing a gas market.	South Africa's potential for future gas demand will be driven by four key sectors such as power, synfuels, industry and transport.	A gas market with domestic partners will curb gas-price volatility caused by the international markets. Therefore, an established gas market in South Africa will ensure accessibility and affordability.
36.	What do you think is the optimal timelines within which this capacity should be made available on the grid, considering the urgency that is required to deal with load shedding in the country?	The targeted COD for the requested capacity would be between the years 2020 to 2028.	- The optimal timelines are as emphasized in the determination which must not be beyond 2028. - In its feasibility study, Eskom indicated indicative timelines of commissioning different units of which all of the 4 units will be commissioned by no later than the end of year 2028.
ii. COST RECOVERY MECHANISM AND THE IMPACT ON TARIFFS			
1.	Should NERSA continue with the current cost recovery mechanism stipulated in section 13 of	The Stakeholder did not answer this question directly.	

	the current Multi-Year Price Determination Methodology (MYPD) and section 14 of NERSA's CRM Rules? If not, please propose the most suitable cost recovery mechanism for the gas project that would ensure that investment in a gas-to-power facility is economic, resulting in a good opportunity cost for the investor over the life of the asset, while also ensuring that electricity remains affordable.		
2.	Provide your thoughts on the costs that will be associated with establishing the new allocated generation capacity in line with the mandate to ensure the longterm sustainability of the electricity supply industry, as well as affordability and equity.	• No comment was provided.	• EPRI and Lazard provide the overnight cost of 3 000MW produced from gas. Although these costs are indicative but they provide an indication how much it will cost to produce 3 000MW from gas.
3.	How should the cost associated with the fuel for this technology be dealt with within the cost recovery mechanism in order to mitigate against	• Natural gas should be aggregated at a national level by State Owned Entities.	• The CRA or any other regulatory price methodology that will be available at that time will be used to mitigate against rapid changes in prices of gas.

	rapid changes in prices of gas?		
4.	What will be the anticipated tariff impact to customers or end users in R/kWh if 3 000MW of gas-to-power is procured?	No comment was provided.	The revenue requirement or any other price determination methodology that will be available at that time, will be used to calculate the tariff impact of 3 000MW.
5.	Provide your comments on the socio-economic impact of the determined capacity in terms of the number of jobs that can be created, including the objective of a just transition from coal to clean technologies.	Hundreds to thousands of jobs will be created during the construction phase of the project, which will create those jobs for 3 to 5 years.	- Since the Electricity Supply Industry (ESI) is transitioning in line with Just Energy Transition, the 3 000MW from gas technology will minimize the impact of Eskom staff members who might lose their jobs because of some of the coal fired power stations reaching the end of their useful lives. - It will also create jobs to the locals during different stages of project life cycle.
iii. ENVIRONMENTAL IMPACT OF THE NEW GENERATION CAPACITY OF 3 000 MW FROM GAS AT RICHARDS BAY			
1.	Do you think that it is proper for the Energy Regulator to deviate from its mandate and consider other Government Departments mandates when concluding section 34 determinations?	No, each Government Department should execute its mandate.	This determination derives its mandate from the Electricity Regulations Act. NERSA does not have powers over mandates of other Government Departments.
2.	Is it proper for the Energy Regulator to consider matters that will be dealt	No, following the similar reasoning provided in 1.	This determination derives its mandate from the Electricity Regulations Act. NERSA does not

	with at licensing stage during the concurrence process?		have powers over mandates of other Government Departments.
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IV. PROCUREMENT PROCESS UNDER THE IPP PROCUREMENT PROGRAMME

1.	Provide your thoughts on Eskom as the chosen procurer of this new generation capacity.	It is logical for Eskom being chosen as a procurer of this new generation capacity .	The chosen Procurer falls outside of the NERSA mandate.
2.	Provide your thoughts on the method chosen for the procurement of the new generation capacity.	There should be an appointed procuring agency that will make sure that procurement is done in a manner where the permit related tasks for the complete gas-supply chain are completed before the actual project construction to avoid project delays.	Procurement process falls outside of the NERSA mandate. These powers are for other Government Departments to execute.
3.	Section 5 of the Section 34 determination states that the procurement of primary energy and its associated logistic infrastructure must be done in a manner supportive of the State's participation through other State Owned Companies that have the relevant mandate(s). Provide your thoughts on this requirement.	It would be very helpful if the counterparty on each procurement process is a SOE, instead of the SOEs actually competing with private sector to complete projects for each process.	This falls outside of the Regulator's mandate. However, Public (SOEs)-Private-Partnerships (PPPs) are known for producing most desirable outcomes in building mega projects.

V. CONCURRENCE PROCESS TO THE MINISTERIAL DETERMINATIONS WITH INDICATIVE TIMELINES

10. RICHARDS BAY IDZ

No.	Questions from the consultation paper	Stakeholder Comment	Reason for Not Supporting
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i. TECHNOLOGY			
1.	In your view, is gas the type of technology that is in line with best practice and will the determined capacity ensure security of supply?	We fully agree that gas is the appropriate technology that is in line with best practice. We do not however agree that the 3,000 MW proposed for Eskom will be sufficient and believe that, in addition to the 3,000 MW for which Eskom is seeking a determination, at least the further 3,000 MW contemplated in the Integrated Resources Plan ("IRP") should be awarded to Independent Power Producers, including Independent Power Producers in Richards Bay.	• The 3 000MW to be generated from Gas technology is in line with best practice because it will improve the supply-demand imbalance in the country and support the Electricity Regulations Act's objective to promote the use of diverse energy sources and energy efficiency. • It will also support the Government's initiatives to bring affordable, reliable and cleaner energy to power consumers. • However, this analysis provided above does not take away the fact that this is not a section 34 determination. Therefore, another process to facilitate this application should be followed
2.	Provide your comments on the availability of natural gas in South Africa	Subject to what is said below regarding the risks posed by the environmental lobby, the supply of gas to South Africa is looking positive. Sasol, which currently uses approximately 110 PJ of gas and is looking to add an additional 40-60 PJ to replace coal in order to reduce its carbon footprint. It is looking for LNG imports from Mozambique and other suppliers and has indicated: <i>"We are looking at two sources of LNG importation: Matola and Richards Bay. In terms of Matola, we are advanced. In terms of Richards Bay, it is going to take a</i>	• The gas availability in South Africa is a concern. However, Eskom is in a process of engaging with partners like Transnet whose role will be to participate in Liquefied Natural Gas (LNG) receiving infrastructure, regasification, gas transmission and distribution.

		<i>while. We are working jointly with the CEF, as well as government, to see whether we can accelerate the developments. We need the two options to ensure security of supply and flexibility.”³⁸(our emphasis)</i> On 14 September 2022 Transnet issued a Request for Proposal for an Liquefied Natural Gas (“LNG”) terminal in Richards Bay and has already issued a Request for Information, to which it received some 19 responses, indicating a market interest in the importation of gas. Transnet Pipelines has issued a Request for Proposal for partners to jointly tender with it for the LNG Terminal tender. As a back-up, Gigajoule has indicated that it intends to locate a Floating Storage and Regasification Unit (“FSRU”) at Matola in Mozambique which will pump gas through the Rompco pipeline to Secunda. Some have also postulated connecting the Rompco line to the Lilly line. This is however, in our opinion, a second choice, although both Gigajoule and Transnet have indicated that they see their projects as complementary⁴², something Sasol also seems to indicate in the quote above.	
3.	Is the determined capacity sufficient to ensure uninterrupted supply of electricity?	No, we believe that it is not sufficient and that more will be needed, particularly reliable, dispatchable power like gas, which is not dependent on the time of day or weather. See 1.1 above where it is indicated that Eskom itself has indicated that 3,000MW to 6,000MW is needed immediately. We should here add that, in addition to Eskom, there are additional four independent power producers which have been approved to locate in the RBIDZ. There are an additional 2 Independent power producers which have	• The electricity system has been a constraint for most of the past 14 years. There is also a fundamental change in the energy mix towards renewable resources. Therefore, a balanced mid-merit Combined Cycle Power Plant is a technically feasible solution to help with the power crisis.

		conducted environmental approval processes to build land-based gas power plants outside the RBIDZ.	
4.	Provide what you consider to be the risks and challenges associated with this new capacity.	We believe that the two key risks associated with the new capacity are ensuring that the new import facility at the Port of Richards Bay is available when needed and legal challenges by environmental lobby groups. The first risk is being addressed by Transnet, which has issued a Request for Information for the proposed gas import terminal and has indicated that it intends issuing a Request for Proposal in the near future. It is essential that the gas import facility is available on time to feed the power plant. Associated with this is the availability of a FSRU given the scramble for FSRU's as a result on the need for Europe to import LNG in order to replace Russian Gas61. Even here there is an opportunity as, were the war in Ukraine to end suddenly, there would be a glut of FSRU's on the market as they would no longer be needed in Europe, It should be added that the three years needed to build a gas power plant is also sufficient time to build an FSRU. In the alternative, on-shore regasification could be considered. In our view the greater risk is posed by environmental groups, which appear to be opposed to natural gas on principle and appear to have deep enough pockets to delay any gas projects almost indefinitely using litigation.	• The risk of the availability of gas can be minimized if Eskom can find local partners to supply reliable gas, at a required pressure and a relatively stable price to avoid volatility of gas price. Hence, domestic partners can curb international gas volatility. • Also, gas market aggregation can mitigate this risk whereby a lot of players can compete or use bargaining power to buy it cheaper.
5.	Provide your comments on whether the country will benefit more optimally from gas and the benefits of establishing a gas market.	IGUA-SA has cautioned that South Africa will begin facing gas shortages from 2023 onwards and that the yearly shortfall could be as large as 98 PJ from 2025 onwards. Sasol's central processing facility in Mozambique is currently able to produce at a yearly rate of 197 PJ, of which some 110 PJ is consumed by Sasol itself. Today, South Africa consumes approximately (~) 180 Petajoules	• A gas market with domestic partners will curb gas-price volatility caused by the international markets. Therefore, an established gas market in South Africa will ensure accessibility and affordability.

		per annum (PJ/a) of gas, predominantly in the synfuels sector (110PJ/a) and the industrial sector (70PJ/a), which supports up to 56thousand jobs across the value chain, generates up to R215billion (\$14,195,271,800) in taxable revenue and contributes~1–2% of GDP. While it can be argued that the needs of existing and potential gas users can be met from an FSRU at Matola, it must be remembered that: 400,000potential jobs. The plant is thus desperately needed so that investment can create jobs. The country already has load-shedding, against a background where it has very low growth. The construction on the plant needs to begin as soon as possible.	
6.	What do you think is the optimal timelines within which this capacity should be made available on the grid, considering the urgency that is required to deal with load shedding in the country?	The construction of the plant needs to begin as soon as possible.	• The optimal timelines are as emphasized in the determination which must not be beyond 2028. • In its feasibility study, Eskom indicated indicative timelines of commissioning different units of which all of the 4 units will be commissioned by no later than the end of year 2028.
ii. COST RECOVERY MECHANISM AND THE IMPACT ON TARIFFS			
1.	Should NERSA continue with the current cost recovery mechanism stipulated in section 13 of the current Multi-Year	While a precedent may be established, it is suggested that, as long as it was ensured that the costs were properly controlled, pricing on a cost-plus basis would give Eskom the security it needs to obtain funding for the	• Cost Recovery Mechanism should have not been included as part of section 34 determination as it falls

	Price Determination Methodology (MYPD) and section 14 of NERSA's CRM Rules? If not, please propose the most suitable cost recovery mechanism for the gas project that would ensure that investment in a gas-to-power facility is economic, resulting in a good opportunity cost for the investor over the life of the asset, while also ensuring that electricity remains affordable.	project and ensure that electricity was provided at the lowest cost.	within the scope of the MYPD4 Methodology and CRM Rules.
2.	Provide your thoughts on the costs that will be associated with establishing the new allocated generation capacity in line with the mandate to ensure the longterm sustainability of the electricity supply industry, as well as affordability and equity.	No clear answer provided	• EPRI and Lazard provide the overnight cost of 3 000MW produced from gas. Although these costs are indicative but they provide an indication how much it will cost to produce 3 000MW from gas.
3.	How should the cost associated with the fuel for this technology be dealt with within the cost recovery mechanism in order to mitigate against rapid changes in prices of gas?	The operator should be encouraged to enter into long-term agreement for the supply of LNG Gas instead of buying on the spot market.	• The CRA or any other regulatory price methodology that will be available at that time will be used to mitigate against rapid changes in prices of gas.

4.	What will be the anticipated tariff impact to customers or end users in R/kWh if 3 000MW of gas-to-power is procured?	No comment provided	The revenue requirement or any other price determination methodology that will be available at that time, will be used to calculate the tariff impact of 3 000MW.
5.	Provide your comments on the socio-economic impact of the determined capacity in terms of the number of jobs that can be created, including the objective of a just transition from coal to clean technologies.	The only way to create additional jobs is if investors establish new businesses or expand existing businesses. The lack and unpredictability of electricity deters investment and inhibits the growth needed. A more stable electricity supply would improve investment and help create employment. Consequently the proposed gas power plant must not only be considered against the jobs it would create and the jobs it would save, but against the opportunity cost of jobs in the wider economy that would be created if more, and more reliable, electricity was provided and investors were not deterred by the electricity problems.	- Since the Electricity Supply Industry (ESI) is transitioning in line with Just Energy Transition, the 3 000MW from gas technology will minimize the impact of Eskom staff members who might lose their jobs because of some of the coal fired power stations reaching the end of their useful lives. - It will also create jobs to the locals during different stages of project life cycle.
iii. ENVIRONMENTAL IMPACT OF THE NEW GENERATION CAPACITY OF 3 000 MW FROM GAS AT RICHARDS BAY			
1.	Do you think that it is proper for the Energy Regulator to deviate from its mandate and consider other Government Departments mandates when concluding section 34 determinations?	We believe that, in accordance with the Electricity Regulation Act and the Promotion of Administrative Justice Act, NERSA should keep to its legislative mandate. At most, where there are competing jurisdiction issues, it can agree to make its decision conditional on the findings of the Government Department concerned.	This determination derives its mandate from the Electricity Regulations Act. NERSA does not have powers over mandates of other Government Departments.
2.	Is it proper for the Energy Regulator to consider matters that will be dealt with at licensing stage	We believe that the matters in section 34 of the ERA should be considered:	This determination derives its mandate from the Electricity Regulations Act. NERSA does not

	during the concurrence process?		have powers over mandates of other Government Departments.
IV. PROCUREMENT PROCESS UNDER THE IPP PROCUREMENT PROGRAMME			
1.	Provide your thoughts on Eskom as the chosen procurer of this new generation capacity.	The RBIDZ fully supports Eskom as the chosen procurer of this 3000MW of Gas in collaboration with Transnet.	• The chosen Procurer falls outside of the NERSA mandate.
2.	Provide your thoughts on the method chosen for the procurement of the new generation capacity.	We fully agree on the issue of the determination to Eskom as a way to get power into the grid in the shortest possible time, especially given the work that Eskom has already done on the project, including the obtaining of environmental approvals for both the gas power plant and power lines.	• The chosen procurement process falls outside of the NERSA mandate.
3.	Section 5 of the Section 34 determination states that the procurement of primary energy and its associated logistic infrastructure must be done in a manner supportive of the State's participation through other State Owned Companies that have the relevant mandate(s). Provide your thoughts on this requirement.	This ties in with the requirement that Transnet must provide for the port infrastructure needed for the supply of gas. The RBIDZ is well aware that the operation of the power plant was not possible without the co-operation of the port and that, similarly, the creation of port infrastructure for the importation of LNG would require an off-taker. The RBIDZ has consequently sought to align the activities of Transnet and Eskom in this regard, especially after there was no progress on the development of the gas industry following the publication of the Project Information Manual for the development of the gas industry in 2016.	• The chosen procurement process and its logistics fall outside of the NERSA mandate.

11. SASOL

No.	Questions from the consultation paper	Stakeholder Comment	NERSA ANALYSIS ON STKEHOLDER COMMENT
i. TECHNOLOGY			
37.	In your view, is gas the type of technology that is in line with best practice and will the determined capacity ensure security of supply?	• Yes - only if used appropriately and gas as a transition fuel is required to additionally, support non-dispatchable technology such as renewables. It also plays a key role as an alternative to coal as baseload. • The load factor of gas fired generation should be determined by the fuel cost. It may start off at a high load factor to assist with closing the energy gap and can later switch to a low load factor to provide capacity when needed.	• The 3 000MW to be generated from Gas technology is in line with best practice because it will improve the supply-demand imbalance in the country and support the Electricity Regulations Act's objective to promote the use of diverse energy sources and energy efficiency. • It will also support the Government's initiatives to bring affordable, reliable and cleaner energy to power consumers.
38.	Provide your comments on the availability of natural gas in South Africa	• Natural gas is only available in limited quantities in South Africa. The main current source of gas is Mozambique. This resource is depleting, with no clear opportunities for replacement. LNG is currently the most probable source of the gas required to enable this GtP development.	• The gas availability in South Africa is a concern. However, Eskom is in a process of engaging with partners like Transnet whose role will be to participate in Liquefied Natural Gas (LNG) receiving infrastructure, regasification, gas transmission and distribution.
39.	Is the determined capacity sufficient to ensure uninterrupted supply of electricity?	• The determined capacity is not insufficient but will assist in closing the gap however substantial additional capacity is required.	• The electricity system has been constraint for most of the past 14 years. There is also a fundamental

			change in the energy mix towards renewable resources. Therefore, balanced mid-merit Combined Cycle Power Plant is a technically feasible solution to help with power crisis.
40.	Provide what you consider to be the risks and challenges associated with this new capacity.	• Risks: Exposure to LNG price and Rate of Exchange. • Litigation from environmental groups fundamentally opposed to all fossil fuels and strict environmental requirements on fossil fuels and uncertainty on future litigation measures.	• The risk of the availability of gas can be minimized if Eskom can find local partners to supply reliable gas, at a required pressure and a relatively stable price to avoid volatility of gas price. Hence, domestic partners can curb international gas volatility. • Also, gas market aggregation can mitigate this risk whereby a lot of players can compete or use bargaining power to buy it cheaper.
41.	Provide your comments on whether the country will benefit more optimally from gas and the benefits of establishing a gas market.	• Increasing gas supply ensures that the transition to greener options is done in a just manner and allows for economic growth through increased electricity supply and additional industrial demand.	• A gas market with domestic partners will curb gas-price volatility caused by the international markets. Therefore, an established gas market in South Africa will ensure accessibility and affordability.
42.	What do you think is the optimal timelines within which this capacity should be made available on the grid, considering the	• An integrated permitting process between supply infrastructure and gas to power can see the first terminal materialize as soon as possible (~2027-2028). This will	• The optimal timelines are as emphasized in the determination which must not be beyond 2028.

	urgency that is required to deal with load shedding in the country?	also hopefully move out of the current heated LNG market dynamics.	• In its feasibility study, Eskom indicated indicative timelines of commissioning different units of which all of the 4 units will be commissioned by no later than the end of year 2028.
ii. COST RECOVERY MECHANISM AND THE IMPACT ON TARIFFS			
1.	Should NERSA continue with the current cost recovery mechanism stipulated in section 13 of the current Multi-Year Price Determination Methodology (MYPD) and section 14 of NERSA's CRM Rules? If not, please propose the most suitable cost recovery mechanism for the gas project that would ensure that investment in a gas-to-power facility is economic, resulting in a good opportunity cost for the investor over the life of the asset, while also ensuring that electricity remains affordable.	• No - the current methodology does not allow operators to recover costs in a timely manner. At best this would lead to expensive hedging strategies being built into tariff applications and at worst the independent power producer (IPP) might not survive. Both outcomes are ultimately bad for the consumer and SA in general.	• Cost Recovery Mechanism should have not been included as part of section 34 determination as it falls within the scope of the MYPD4 Methodology and CRM Rules.
2.	Provide your thoughts on the costs that will be associated with establishing the new allocated generation capacity in line with the	• The stakeholder did not provide any costs associated with establishing 3 000MW from gas technology.	• EPRI and Lazard provide the overnight cost of 3 000MW produced from gas. Although these costs are indicative but they provide an

	mandate to ensure the longterm sustainability of the electricity supply industry, as well as affordability and equity.		indication how much it will cost to produce 3 000MW from gas.
3.	How should the cost associated with the fuel for this technology be dealt with within the cost recovery mechanism in order to mitigate against rapid changes in prices of gas?	Fuel cost must be a pass-through, with an appropriate fuel sourcing strategy to protect customers from spot market volatility.	The CRA or any other regulatory price methodology that will be available at that time will be used to mitigate against rapid changes in prices of gas.
4.	What will be the anticipated tariff impact to customers or end users in R/kWh if 3 000MW of gas-to-power is procured?	No comments provided on this question.	The revenue requirement or any other price determination methodology that will be available at that time, will be used to calculate the tariff impact of 3 000MW.
5.	Provide your comments on the socio-economic impact of the determined capacity in terms of the number of jobs that can be created, including the objective of a just transition from coal to clean technologies.	Building GtP including the associated LNG value chain has the potential to create many permanent construction jobs (as demonstrated by ROMPCO).	- Since the Electricity Supply Industry (ESI) is transitioning in line with Just Energy Transition, the 3 000MW from gas technology will minimize the impact of Eskom staff members who might lose their jobs because of some of the coal fired power stations reaching the end of their useful lives. - It will also create jobs to the locals during different stages of project life cycle.
iii. ENVIRONMENTAL IMPACT OF THE NEW GENERATION CAPACITY OF 3 000 MW FROM GAS AT RICHARDS BAY			

1.	Do you think that it is proper for the Energy Regulator to deviate from its mandate and consider other Government Departments mandates when concluding section 34 determinations?	• NERSA cannot make decisions that impact other departments in isolation.	• This determination derives its mandate from the Electricity Regulations Act. NERSA does not have powers over mandates of other Government Departments.
2.	Is it proper for the Energy Regulator to consider matters that will be dealt with at licensing stage during the concurrence process?	• Fore sighting of licencing matters during the concurrence process would be advantageous as it will contribute towards streamlined and seamless implementation of the determination.	• This determination derives its mandate from the Electricity Regulations Act. NERSA does not have powers over mandates of other Government Departments.
IV. PROCUREMENT PROCESS UNDER THE IPP PROCUREMENT PROGRAMME			
1.	Provide your thoughts on Eskom as the chosen procurer of this new generation capacity.	• Eskom is currently under extreme financial pressure and may not be in a position to execute on the directive.	• The chosen Procurer falls outside of the NERSA mandate.
2.	Provide your thoughts on the method chosen for the procurement of the new generation capacity.	• The IPP process appears to be a fair way of procuring be generation capacity and should be followed.	• The chosen procurement method falls outside of the NERSA mandate.
3.	Section 5 of the Section 34 determination states that the procurement of primary energy and its associated logistic infrastructure must be done in a manner supportive of the State's participation through other State Owned Companies that have the	• State-owned entities (SoEs) should compete with interested parties on an equal footing. The determination as written appears to pre-determine at least part participation, making bidding processes opaque and may deter some interest.	• This falls outside of the Regulator's mandate. However, Public (SOEs)-Private-Partnerships (PPPs) are known for producing most desirable outcomes in building mega projects.

	relevant mandate(s). Provide your thoughts on this requirement.		
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